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ZEMA/INS/101/04/1

In reply please quote

August 16, 2024.

No:

The Council Secretary
Chirundu Town Council,
CHIRUNDU.

Dear Sir,

REF: REQUEST FOR COMMENTS ON THE PROPOSED MINERAL EXPLORATION PROJECT ON LICENSE NO. 36411-HQ-LEL IN CHIEFTAINNESS SIKOONGO'S AREA, CHIRUNDU DISTRICT BY GRAND PROPEROUS COMPANY LIMITED.

The Zambia Environmental Management Agency (ZEMA) is in receipt of the Environmental Project Brief (EPB) for the above captioned project. The submission has been made in accordance with the provisions of the Environmental Management Act No.12 of 2011 and the Environmental Impact Assessment Regulations No. 28 of 1997. ZEMA is expected to review the EPB and issue a decision. The review process is supposed to take into account the concerns of all stakeholders. It is for this reason that a copy of the said EPB is being sent to you as a key stakeholder to facilitate your input in the review process. Specifically, we wish to find out if the project is in **compliance with your legal, institutional and management requirements.**

You may further wish to respond to or comment on the following key questions:

1. Is the project purpose and objective justifiable i.e. is the project acceptable?
2. Is the proposed project location suitable for the purpose?
3. Are the proposed mitigation and enhancement measures adequate or can they be improved upon?

By law you are required to submit your comments within thirty days; however, we would appreciate your written response before **September 16, 2024**. This is in order for ZEMA to incorporate your comments/concerns into the final decision.

Please be informed that if we do not receive your comments by the said date, the Agency will proceed to make a decision.

We look forward to your support.

Yours faithfully,

Samuel Daka
Senior Inspector
For/Manager – Environmental Assessments
For/Director General
ZAMBIA ENVIRONMENTAL MANAGEMENT AGENCY.

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GRAND PROSPEROUS COMPANY LIMITED

Proposed Mineral Exploration Project on 36411-HQ-LEL in Chieftainess Sikoongo's Area, Chirundu District, Southern Province

Environmental Project Brief

Prepared by



July 2024

EXECUTIVE SUMMARY

Introduction

Grand Prosperous Company Limited ("Grand Prosperous"), a company registered in Zambia in 2015, has acquired a large-scale exploration license 36411-HQ-LEL in Chirundu District of Southern Province and intends to carry out exploration activities to ascertain the availability and quantity of the minerals in the licence area. The licence area covers an area of about 2436.84 hectares. The company will be exploring for Feldspar, Gypsum, Talc, Quartz, Mica, Lithium and Florite.

Project Objective

The objective of the proposed project by Grand Prosperous is to conduct exploration for Feldspar, Gypsum, Talc, Quartz, Mica, Lithium and Florite on its exploration licence area 36411-HQ-LEL.

Project Location and Access

The proposed project, exploration licence 36411-HQ-LEL, is located in HRH Chief Sikoongo's chiefdom in Lusitu area of Chirundu District. Access to the site is from the Siavonga (M15) Road at Lusitu, 5km before the Lusitu River, heading west for some 4km up to the eastern boundary of the project license.

Project Cost

The project's investment cost is USD50,000.00.

Legislative and Institutional Framework

The following policies, institutional framework and pieces of legislation were reviewed.

Legislative Framework

The following are some of the pieces of legislation related and relevant to the project being developed by Grand Prosperous:

- The Environmental Management Act No. 12 of 2011 as amended by Environmental Management (Amendment) Act No. 8 of 2023
- The Mines and Minerals Development Act (MMDA) No. 11 of 2015
- The Environmental Protection and Pollution Control (Environmental Impact Assessment) Regulations, SI 28 of 1997
- The Mines and Minerals Development Act (MMDA) No. 11 of 2015.
- Statutory Instrument No 29 of 1997, which is the Mines and Minerals (Environmental) Regulations.
- The Environment Management Act (Licensing) Regulations (SI 112 of 2013)
- The Urban and Regional Planning Act, 2015

- The Petroleum Act, No. 13 of 1994
- The Local Government Act, No. 2 of 2019
- The Pneumoconiosis Act No. 13 of 1994.
- Ionising Radiation Protection Act No. 16 of 2005.
- The Employment Act, 2015
- The Workers Compensation Act, No. 10 of 1999
- The Occupational Health and Safety Act, No. 36 of 2010
- The Public Health Act, 1930
- The Factories Act, 1999
- The National Heritage Conservation Commission (NHCC) Act, 1986

Institutional Framework

Institutions relevant to the development of the water project include the following:

- Zambia Environmental Management Agency
- Ministry of Mines and Minerals Development
- Chirundu Town Council

Project Description

Grand Prosperous proposes to conduct exploration on exploration licence area 36411-HQ-LEL of Chirundu in Chieftainess Sikoongo's Lusitu area. Grand Prosperous now intends to carry out exploration activities to ascertain the availability and quantity of the minerals in the licence area. The licence area covers an area of about 17,375.938 hectares. The company will be exploring for copper and gold. The development will go through the following activities:

Construction Phase

- The construction phase will involve desk study which will mainly involve review of available literature for the licence area; and site preparation.
- The areas that will be cleared will include drilling sites, sampling sites, access roads and camping areas.

Operation Phase

This phase will include:

- Geochemical sampling: - This will involve sampling of geological materials such as soils, stream sediments, rocks and biological materials for geochemical analyses. These activities will be followed by plotting of the results on a map and interpretation of results;
- Reverse Circulation Drilling (RCD): - This utilizes a solid bit to drill a hole and deliver rock chips to the surface. RCD activities are all automobile mounted hence the need for access roads; and

- **Diamond Drilling:** - This method involves use of hollow drilling bits that are impregnated with diamond chips. As the bit rotates, it cuts a circular portion of the rock which is pushed up into the core barrel of the drill. The rock core is what is used for assaying.
- **Bulk sampling:** - This will involve sampling of mineralized geological materials such as soils and rocks for mineral content analyses. These activities will be followed by plotting of the results on a map and generation of an orebody which will determine the appropriate mining method for the mineral resource.

Decommissioning Phase

At decommissioning and closure stage the exploration site will be dismantled, drilling pits/trenches buried and closure of the camp site. The usable materials will be identified, sorted and reused/recycles or sold off.

Analysis of the Project Alternatives

The No Option Alternative

The "No Project Option" was considered. This option was not selected because it would negatively affect the development of the mineral resource and hence deprive the country of the much-needed foreign exchange that the project would earn through the export of minerals if the results shall be good and mining is embarked on. This option would also deprive the local people of the much-needed employment opportunity.

Site Alternative

Most of the potential areas in Zambia are under exploration activities hence the choice of where to conduct exploration is largely determined by availability and issuance of licences by the Ministry of Mines and Minerals Development. Alternative areas could not be considered because the licence is site specific.

Technology

Exploration method alternatives that were considered are soil sampling, sediment sampling, trenching and drilling. All these alternatives shall be used because the combination of these exploration methods gives the best exploration results.

The alternatives of technology that were considered for soil sampling, sediment sampling and trenching are hand-held picks and shovels against the use of an excavator. Hand-held picks and shovels were preferred because they make smaller excavations and hence are environmentally friendly as compared to the excavator.

Environmental Impacts and Mitigation Measures

The potential impacts are discussed according to how they relate to the proposed project components and its associated infrastructure. Below are the significant impacts of the proposed project.

Impacts and mitigation measures

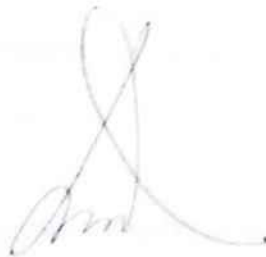
Impact	Enhancement/Mitigation measure
Employment opportunities created	Sino will offer employment opportunities to the local people
Business opportunities created	Offer the supply of goods and services to local businesses
Enhanced local and national economy	Prompt remittance of pay-as-you earn, duties and taxes
Noise pollution from machinery	- Regularly maintain all vehicles and equipment to ensure that they are mechanically sound. - Ensure that where workers are exposed to elevated sound levels, use of PPE such as ear muffs/plugs will be mandatory. - limit exploration to daytime only; and - good maintenance and proper operation of machinery to minimize noise generation.
Generation of waste	Designate waste receptacles at identified points for temporal storage before disposal
Impact due to Occupational Health Concerns	a) Provision of personal protective equipment (PPE) to workers; b) Provision of a well-stocked first aid box;
Fire Risks	a) Procuring enough fire-fighting equipment such as fire extinguishers that will be serviced from time to time; b) Securing of all electrical equipment; c) Training personnel in first aid, fire-fighting, and emergence response procedures. d) Grand Prosperous will make provisions to avoid interfering with natural fire regimes.
Impact of HIV/AIDS	Education and sensitization on the dangers of HIV/AIDS together with the promotion of self-protection e.g. by use of condoms will be the key intervention.

Environmental Management Plan

In order to effectively manage the environmental impacts and the proposed mitigation measures identified above, Grand Prosperous has developed an Environmental Management Plan (EMP).

Conclusion and Recommendations

Even though a number of environmental impacts have been identified, management of Grand Prosperous commits itself to adhere to the proposed mitigation measures in this EPB. Management is committed to comply with the relevant pieces of legislation identified in this EPB. The environmental management and monitoring plans provided in this document will be used by Grand Prosperous for the systematic management of significant environmental aspects.



Director – Grand Prosperous Company Limited

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ABBREVIATIONS

CBD	Central Business District
CTC	Chirundu Town Council
EIA	Environmental Impact Assessment
EMA	Environmental Management Act
EPB	Environmental Project Brief
ERB	Energy Regulation Board
GRZ	Government of the Republic of Zambia
HCF	Health Care Facility
HQ	Head Quarters
IAPs	Interested and Affected Parties
Kg	Kilogram
L	Litre
LDC	Least Developed Country
LEL	Large Scale Exploration License
m ²	Square Meter
m ³	Cubic Meter
NHCC	National Heritage Conservation Commission
RDA	Roads Development Agency
RPA	Radiation Protection Agency
PPE	Personal Protective Equipment
SDG	Sustainable Development Goal
SI	Statutory Instrument
US\$	United States Dollar
ZABS	Zambia Bureau of Standards
ZEMA	Zambian Environmental Management Agency

1 INTRODUCTION

1.1 PROJECT BACKGROUND AND RATIONALE

Grand Prosperous Company Limited ("Grand Prosperous"), a company registered in Zambia in 2015, has been formed to explore mineral deposits in Zambia and has acquired a large-scale exploration license 36411-HQ-LEL in Chirundu District of Southern Province. Grand Prosperous intends to carry out exploration activities to ascertain the availability and quantity of the minerals in the licence area. The licence area covers an area of about 2436.84 hectares. The company will be exploring for Feldspar, Gypsum, Talc, Quartz, Mica, Lithium and Florite.

1.2 PROJECT OBJECTIVE

The objective of the proposed project by Grand Prosperous is to conduct mineral exploration for Feldspar, Gypsum, Talc, Quartz, Mica, Lithium and Florite on its exploration licence area 36411-HQ-LEL (Appendix 2) in HRH Chieftainess Sikoongo's area in Chirundu District of Southern Province.

Depending on the results, mining activities shall be embarked on and these activities shall undergo separate Environmental Impact Assessment as this EPB only covers exploration activities.

1.3 PROJECT DIRECTORS AND SHAREHOLDERS

The Shareholders of Grand Prosperous are indicated in the list below in Table 1-1.

Table 1-1: Grand Prosperous Company Shareholders

Shareholder	Nationality	% Shareholding
Xiaoming Liang	Chinese	67.87
Yuye Liu	Chinese	25.47
Scholasticah Chali	Zambian	6.66

1.4 CONTACT DETAILS

The company can be contacted using the details indicated below:

Table 1-2: Contact Details

Physical Address	Phone & Email
Plot No. 762, Dallas, Kabwe	Liu Yuye +260 97 7968465 julianliu32@gmail.com

1.5 LOCATION AND ACCESS

The proposed project, exploration licence 36411-HQ-LEL, is located in HRH Chief Sikoongo's chiefdom in Lusitu area of Chirundu District. Access to the site is from the Siavonga (M15) Road at Lusitu, 5km before the Lusitu River, heading west for some 4km up to the eastern boundary of the project license (fig 1-1).

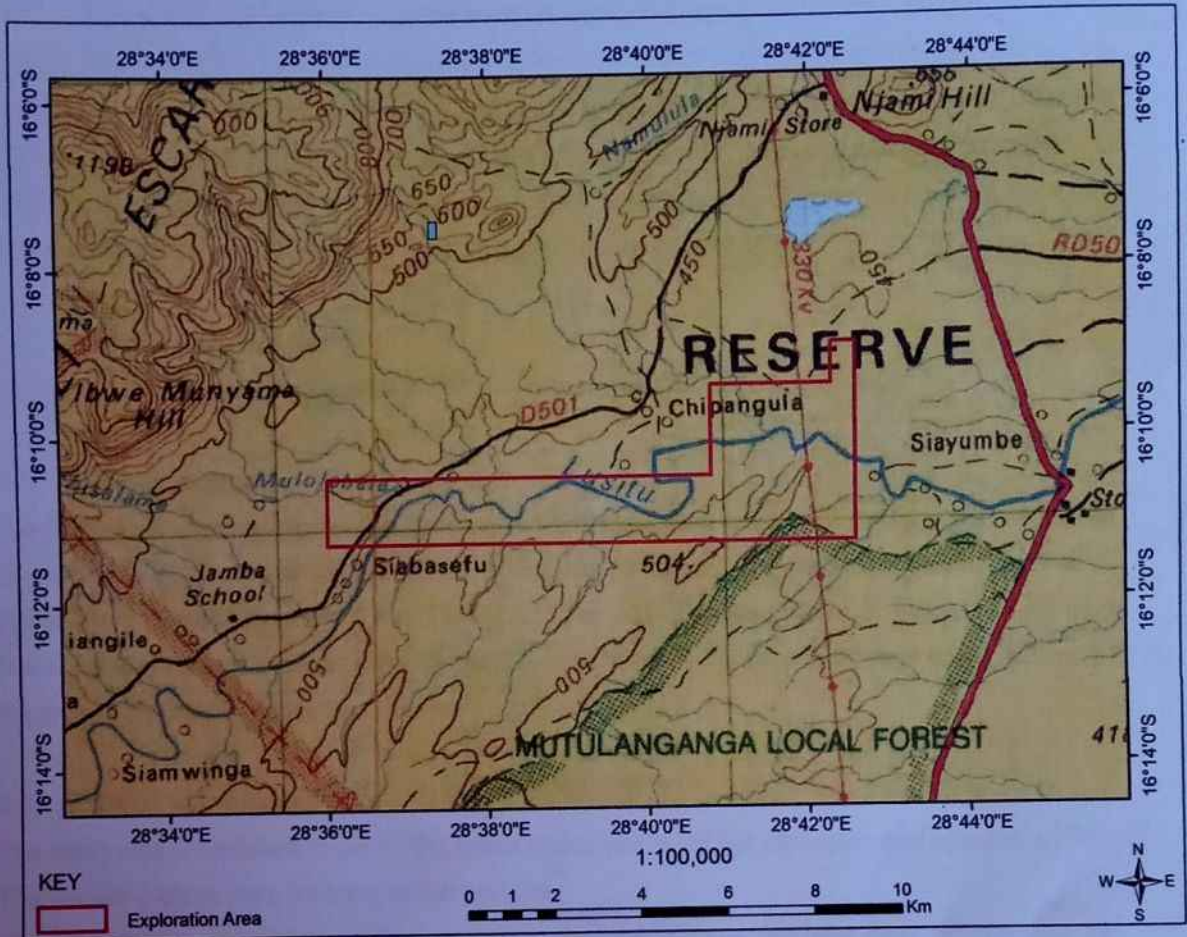


Figure 1-1. Topographic map of the area of interest

The site is bound by the geographic coordinates are indicated in table 1-3 below:

Table 1-3. Coordinates for 36411-HQ-LEL

Southings	Eastings
16° 08' 55"	28° 42' 36"
16° 11' 18"	28° 42' 36"
16° 11' 18"	28° 36' 00"
16° 10' 31"	28° 36' 00"

16° 09' 31"	28° 40' 49"
16° 09' 25"	28° 40' 49"
16° 09' 25"	28° 42' 18"
16° 08' 55"	28° 42' 18"

1.6 TRACK RECORD (PREVIOUS EXPERIENCE OF THE ENTERPRISE)

Grand Prosperous is a Zambian company incorporated in 2015 with interest in investing in mining business. The company now desires to establish a successful mining entity and has therefore embarked on exploration activities for different minerals and various parts of the country to determine which areas will be feasible for further development. Grand Prosperous has been carrying out exploration activities in Southern and Lusaka Provinces.

1.7 COST OF THE PROJECT

The cost of the project is estimated at Fifty Thousand US Dollars (USD50, 000).

1.8 SCOPE OF STUDY

The study for the preparation of this Environmental Project Brief adheres to the requirements of the Environmental Impact Assessment (EIA) regulations (SI 28 of 1997) as well the requirements of the Environmental Management Act (EMA) No. 12 of 2011. The two pieces of legislation demand that socio-economic and environmental impacts consequential to the construction and operation of the project are assessed well in advance.

1.9 STUDY METHODOLOGY

The study was undertaken to obtain the views and concerns of the interested and affected parties. The following steps were involved in the process:

- Consultative discussions with developers of Grand Prosperous and local residents in the immediate environment of the project;
- Review of relevant legislation;
- Review of environmental literature; and
- Identification of significant environmental impacts and development of mitigating measures.

Responses from consultations and data from literature provided the significant environmental impacts and helped in the development of an environmental management and monitoring plan.

1.10 PROPOSED PROJECT IMPLEMENTATION DATE

The project is expected to commence as soon as all the approval requirements are met and done through the various relevant line organizations and institutions which include the EPB approval process by ZEMA. It will take a span of three years to be completed.

The timeline of the project is presented in Table 1-4 presented below.

Table 1-4. Timeline for project implementation

No.	Task Name	Duration	Start	Finish
1.	Preparation	3 months	July 2024	Sep 2024
2.	Operation	36 months	Oct 2024	Sep 2027
3.	Decommissioning and Closure	3 months	Oct 2027	Dec 2027

2 LEGISLATIVE AND INSTITUTIONAL REQUIREMENTS

2.1 LEGAL FRAMEWORK

2.1.1 The Environmental Management Act No. 12 of 2011 as amended by Environmental Management (Amendment) Act No. 8 of 2023

The principal legislation governing environmental management in Zambia is the Environmental Management Act (EMA) of 2011. The Act provides for the sustainable management of natural resources and protection of the environment, and the prevention and control of pollution.

Relevance

Of particular relevance is section 29 of the Act which states that *"A person shall not undertake any project that may have an effect on the environment without the written approval of the Agency, and except in accordance with any conditions imposed in that approval"*.

In relation to this project, some of the functions of ZEMA are to review the environmental impact assessment report, and undertake environmental auditing and monitoring. The act also provides for public participation in decision-making and access to environmental information.

Compliance

This EIA report for the exploration project, in form of an EPB, is being prepared before the construction phase works begin. The reports will be submitted to ZEMA for consideration and approval.

2.1.2 The Environmental Protection and Pollution Control (Environmental Impact Assessment) Regulations, SI 28 of 1997

Part II of the Environmental Impact Assessment (EIA) Regulations, Statutory Instrument 28 of 1997, demands that before a developer commences implementing a project, an environmental project brief (EPB) be prepared and submitted to the relevant regulatory authority for review and approval.

The EIA regulations, Part II (3)(2), specifically require that a developer prepares and submits a project brief for:

- a) *Any project set out in the First Schedule, whether or not the project is part of a previously approved project;*
- b) *Any alterations or extensions of any existing project which is set out in the First Schedule, or;*
- c) *Any project which is not specified in the First Schedule, but for which the Council determines a project brief should be prepared.*

Relevance

According to the EIA Regulations, the proposed project within the category of First Schedule projects and which require that an EPB be prepared and submitted for approval. A project of this nature requires that environmental issues relevant to the project construction and implementation are identified at an early stage and environmental management measures incorporated at the planning stage.

The project proponent is expected to develop an *"impact management plan containing a description of measures proposed for preventing, minimising or compensating for any adverse impact"* of the project.

Compliance

The EIA report is being prepared before the exploration works begin. The impacts associated with exploration are not deemed significant and therefore requires that an EPB be prepared for consideration by ZEMA in line with the First Schedule of the EIA regulations.

2.1.3 The Environment Management Act (Licensing) Regulations (SI 112 of 2013)

There are several parts in this SI giving regulatory powers to ZEMA to control the discharge of water pollutants, air emission pollutants, pesticides and other toxic substances, waste (both non-hazardous municipal and hazardous) and ozone depleting substances. The parts of relevance to the project at hand are II and III. Part II This part of the SI gives powers to ZEMA to regulate discharge of water pollutants and emission of air pollutants into the atmosphere in order to safeguard the general health, safety or welfare of persons, animal life, and plant life.

Relevance

The clearing of activities may generate dust that will need to be managed in compliance with the above regulations. The SI will also govern the discharge of any wastewater that may be produced at any stage of the project.

Compliance

Regular watering down of dusty areas will ensure that the emission limit of 70 µg/m³, for ambient respirable particulate matter (PM₁₀) around the project site, as specified in the Second Schedule, is set within limits.

Table 2-1. Guideline Limit for Ambient Dust

Parameter	Reference Time		Guideline Limit
Respirable particulate matter, PM ₁₀	PM ₁₀	24 hours	70µg/m ³

Source – Second Schedule, SI 112 of 2013

2.1.4 The Mines and Minerals Development Act (MMDA) No. 11 of 2015

The Mines and Minerals Development Act (MMDA) No. 11 of 2015 provides for the duties and powers of officers authorized to enforce the provisions of the Act and its subsidiary Statutory Instruments. This Act applies to the project because it is a mining related project hence all the applicable provisions have to be complied with. These include conducting scheduled activities, appointment of a competent person to oversee mining activities and submission of necessary statutory returns. The subsidiary Statutory Instruments of the MMDA that is relevant to the proposed project are:

- (a) **Statutory Instrument No 29 of 1997, which is the Mines and Minerals (Environmental) Regulations**, provides the framework for conducting and reviewing Environmental Impact Assessment Reports for all mining related projects in the mining sector. Further to this, it provides regulations for auditing project implementation. These Regulations apply to the project because this project will eventually lead to mining activities which will be subjected to a full EIA.

Relevance

Exploration works are part of mining related projects governed and controlled by this piece of legislation.

Compliance

This EPB has identified environmental and social impacts associated with the project and has also proposed mitigation measures supported by an environmental management plan.

- (b) **Mines and Minerals Environmental Protection Fund Regulations** (Statutory Instrument No 102 of 1998) provides the mechanism for setting up and operating the Environmental Protection Fund (EPF).

Relevance

Exploration activities are mining related activities that are expected to contribute to the EPF.

Compliance

Grand Prosperous shall conduct periodic environmental audits, make annual contributions to the EPF and shall conduct progressive site rehabilitation activities.

2.1.5 The Employment Code Act, No. 2 of 2019

This Act provides legislation relating to the employment of persons; to make provision for the engagement of persons on contracts of service and to provide for the form of and enforcement of

contracts of service; to make provision for the appointment of officers of the Labour Department and for the conferring of powers on such officers and upon medical officers; to make provision for the protection of wages of employees; and to provide for the control of employment agencies.

Relevance

The exploration works will involve the employment of twenty-five people.

Compliance

In compliance with this law, Grand Prosperous will ensure that individuals employed are of legal employment age, and are provided with conditions of service that meet or exceed the minimum conditions of service. Employees will not be subjected to exploitation and abuse of their rights through casualisation.

According to this law, it is illegal for any employer to engage an employee on casual basis for any job that is of a permanent nature. The Act also bans unjustified termination of employment by employers

2.1.6 The Public Health Act, No. 22 of 1995

This Act provides for the prevention and suppression of diseases and the general regulation of all matters connected with public health in Zambia. Amongst other things, the Act prohibits anyone from causing a nuisance, where nuisances are given to include:

- The pollution of potable water;
- Any collection of water or any cesspit, latrine or urinal found to contain mosquito larvae;
- Any collection of water, sewage or waste which permits or facilitates the breeding of parasites, insects or other agents which may lead to the infection of people or domestic animals;
- The accumulation or deposit of waste which is offensive or injurious or dangerous to health;
- The discharge or noxious matter or waste water into a water course not approved for the reception of such discharge;
- Premises without sufficient lighting or ventilation;
- Dangerous buildings and overcrowded premises;
- Factories giving rise to smells and effluents which are offensive or dangerous to health.

Relevance

Along with the portable toilets which will be provided, water of adequate quantity and quality will be required for domestic consumption. Waste will be generated by exploration workers as well as from the camp site.

Compliance

Management will ensure that the project site and more importantly the camp site and subsequently the operational areas exhibit high levels of hygiene in order not to subject workers and the general public to public health nuisances. The project site will, at all times, be kept in clean and sanitary conditions.

2.1.7 The Factories Act, 1999

The Act regulates the conditions of employment in factories and other places of work as regards the safety, health and welfare of persons employed therein. The Act also provides for the examination and inspection of certain plant and machinery in order to ensure safety. The Factories Inspectorate under the Ministry of Labour and Social Security is the mandated authority under the Act.

Relevance

All places of work as defined under this act will be deemed a 'factory' where safety, health and general welfare of employees will be required to be upheld.

Compliance

To ensure that the occupational health and safety of employees is upheld during the construction and operational periods, all employees will be trained in health and safety protocols. All employees will be equipped with adequate and appropriate personal protective equipment and the devices, tools and equipment that will be used by the employees will be regularly inspected, maintained and repaired if found to be defective.

2.1.8 The National Heritage Conservation Commission (NHCC) Act, 1986

The NHCC Act provides for the conservation of ancient, cultural and natural heritage, relics and other objects of aesthetic, historical, pre-historical, archaeological or scientific interest.

Relevance

Even though surveys indicated that the project site does not have any artefact of historical or archaeological value, clearing, digging and drilling can potentially lead to chance-finds of sites or relics of historical, archaeological or cultural value. Therefore, any discovery of such artefacts will be reported to NHCC by Grand Prosperous management.

There is a site (fig. 2-1) located 2km north-east of the project site (in green) that is used for ceremonies by the local people. This site is forested and protected by the local people.

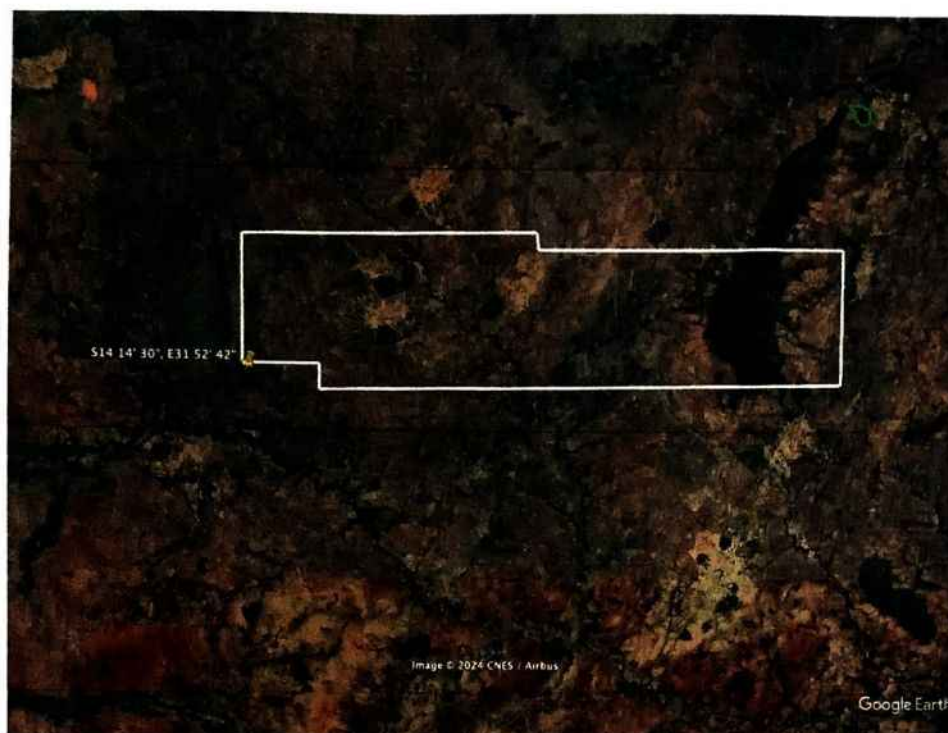


Figure 2-1. Cultural site north-east of the project site

Compliance

In the event that any artefact is found at the project site during exploration by Grand Prosperous, National Heritage and Conservation Commission (NHCC) will be notified. In this case section 37(1) of the Act shall apply which states: *“Any person who desires to excavate any ancient heritage or collect relics shall apply to the Commission for a permit to excavate or collect.”*

2.1.9 The Lands Act, No. 27 of 1995

The Lands Act is the statute governing land administration in Zambia. The law vests all land in the country in the hands of the President and alienation of land by the President.

The Act provides for the alienation, transfer, disposition and change of land. The Act also provides for compulsory acquisition of land by the President whenever he is of the opinion that it is desirable or expedient to do so in the interest of the Republic.

Relevance

The Act gives Zambians an opportunity to participate in the country's economy. The Minister responsible for lands has also issued Land Circular No. 1 of 1985. This Circular laid down detailed procedures to be followed on land alienation.

The Land Act does not comprehensively provide for compensation or resettlement. This is only partially covered in the Mines and Minerals Development Act, Land Survey Act and the Land Acquisition Act.

Compliance

In the acquiring of land, Grand Prosperous has ensured that the right channels are adhered to.

2.1.10 The Local Government Act, No. 2 of 2019

The Act was developed to provide for an integrated local government system; give effect to the decentralisation of functions, responsibilities and services at all levels of local government; ensure democratic participation in, and control of, decision making by the people at the local level; revise the functions of local authorities; provide for the review of tariffs, charges and fees within the area of a local authority; provide for the proceedings of the council and committees; and provide for the role of traditional leadership in democratic governance.

Relevance

The proposed development is within the jurisdiction of Chrirundu District of Southern Province. All approvals required to be sourced from the local authorities will be sought.

Compliance

The contractor and Grand Prosperous will be expected to comply with the specifications and any relevant by laws set up by the local authority.

2.1.11 The Workers Compensation Act, No. 10 of 1999

This Act makes provision for the establishment and administration of a Fund for the compensation of workers disabled by accidents to, or diseases contracted by, such workers in the course of their employment, and for the payment of compensation to dependants of Workers who die as a result of such accidents or diseases; for the payment of contributions to such Fund by employers; for the granting of pensions and allowances to certain dependants of Workers who, being in receipt of pensions for such disablement, die from causes not connected with such accidents or diseases.

Relevance

The construction works and subsequently the operational works through the use of sharp tools and heavy equipment may cause injury to workers.

Compliance

In compliance with this Act, Grand Prosperous and their contractors will make contributions to the Fund on behalf of the workers. Any contracted companies will also be expected to show certificate of contributions made to the Fund.

2.1.12 The Occupational Health and Safety Act, No. 36 of 2010

This Act is promulgated to establish the Occupational Health and Safety Institute and provide for its functions; provides for the establishment of health and safety committees at workplaces and for the health, safety and welfare of persons at work; provide for the duties of manufacturers, importers and suppliers of articles, devices, items and substances for use at work; provide for the protection of persons, other than persons at work, against risks to health or safety arising from, or in connection with, the activities of persons at work.

Section 11 of Part III requires that an employer of ten or more persons at any workplace establishes a health and safety committee.

According to section 13 the functions of the health and safety committee are:

- promote cooperation between the employer and the employees in achieving and maintaining healthy and safe working conditions;
- share information about occupational health, safety and welfare with employees;
- investigate and resolve any matter that may be a risk to the health and safety of employees at a workplace;
- review the measures taken on the health and safety of employees at a workplace; and
- formulate, review and disseminate to the employees the standards, rules and procedures relating to health and safety to be carried out at the workplace.

Relevance

Section 16 of Part IV provides the duties of employers at workplaces in respect of health and safety at workplaces. These duties include:

- ensure, so far as is reasonably practicable, the health, safety and welfare of the employees of the employer at a workplace; and
- place and maintain an employee in an occupational environment adapted to the employee's physical, physiological and psychological ability.

Compliance

To this end, Grand Prosperous and its contractors will ensure that information, instruction, training and supervision are provided to ensure the health and safety of the employees at their workplace.

The duties of an employee at a workplace, according to section 17, with regard to health and safety are to:

- take reasonable care for the employee's own health and safety and that of other persons who may be affected by the employee's acts or omissions at the workplace;
- not operate any machine or engage in a process which is unsafe or is an imminent risk to the employee's own health or safety and that of others; and
- cooperate with the employer or any other person in relation to any duty imposed on the employer or that other person, so far as is necessary to enable that duty or requirement to be performed or complied with.

The employee is also expected to immediately inform the employer, the committee or health and safety representative if there are reasonable grounds to believe that any item, device, article, plant or substance, condition or aspect of the workplace is, or may be, dangerous to the employees' occupational health or safety at or near the workplace.

2.1.13 The Urban and Regional Planning Act, 2 No. 3 of 2015

This piece of legislation was developed, among others reasons, to:

- Provide for a framework for administering and managing urban and regional planning;
- Provide for a planning framework, guidelines, systems and processes for urban and regional planning;
- establish procedures for integrated urban and regional planning in a devolved system of governance so as to ensure multi-sector cooperation, coordination and involvement of different levels of ministries, provincial administration, local authorities, traditional leaders and other stakeholders in urban and regional planning;
- ensure sustainable urban and rural development by promoting environmental, social and economic sustainability in development initiatives and controls at all levels of urban and regional planning.

Relevance

The Act provides for the appointment of regional planning authorities, provincial planning authorities and local planning authorities whose main responsibilities are the preparation, approval of layout plans and revocation of development plans. The project is located within Sinazongwe District of Southern Province.

Compliance

Any development layout plans for the project, will need to be approved by Chirundu Town Council (CTC) in consultation with the Southern Province Planning Authority before project implementation

can commence in accordance with the planning guidelines and framework set by the authorities. The jurisdiction of the Southern Province Planning Authority will be recognised and its specifications will be complied with.

2.1.14 The Gender Equity and Equality Act, No. 22 of 2015

This is an Act to provide for the taking of measures and making of strategic decisions in all spheres of life in order to ensure gender equity, equality and integration of both sexes in society; promote gender equity and equality as a cross cutting issue in all spheres of life and stimulate productive resources and development opportunities for both sexes; prohibit harassment, victimisation and harmful social, cultural and religious practices; provide for public awareness and training on issues of gender equity and equality; and to provide for the elimination of all forms of discrimination against women, empower women and achieve gender equity and equality by giving effect to the Convention on the Elimination of all Forms of Discrimination against Women, the Protocol to the African Charter on Human and People's Rights on the Rights of Women in Africa and the SADC Protocol on Gender and Development.

Relevance

Inequalities in accessing resources and employment opportunities may ensue during project implementation inhibiting economic benefits trickling down to women. The Act is relevant to guide the Project to ensure employment opportunities and recruitment procedures are gender inclusive.

Compliance

To achieve gender equity and equality, the following will be undertaken in line with sections 14 through 20:

- a) both sexes shall enjoy equal rights before the law;
- b) both sexes shall have equal access to justice and protection before the law;
- c) both sexes shall have equal access and rights to training at the workplace;
- d) both sexes shall have equal access to modern, appropriate and affordable technology and support services.

Further to the above,

- i. Grand Prosperous will not discriminate against any sex in the awarding of jobs or provision of jobs and services. Women will have the legal capacity to conclude contracts and other legal instruments.
- ii. Grand Prosperous will not discriminate against a woman due to the woman's pregnancy or the fact that she has given birth to a child.

- iii. Grand Prosperous have a duty to uphold the rights of both sexes and to respect and safeguard the dignity of both sexes and shall not exploit or subject any person to abusive, violent or degrading treatment.
- iv. Grand Prosperous shall:
 - a. develop awareness programmes on fundamental rights and freedoms in order to promote gender equity and equality;
 - b. develop action plans to address gender equity, equality, discrimination and harassment within the public body;
 - c. develop codes of practice and ethics for its officers to assist in promoting gender equity and equality;
 - d. provide assistance, advice and training to its officers on gender equity and equality;
 - e. develop appropriate internal mechanisms to deal with complaints of discrimination and harassment; and
 - f. conduct information and advocacy campaigns to promote gender equity and equality.

2.1.15 The Solid Waste Regulation and Management Act, No. 20 of 2018

This Act was promulgated to provide for the sustainable regulation and management of solid waste; general and self-service solid waste services; the incorporation of solid waste management companies and define their statutory functions; the licensing and functions of solid waste service providers, operators and self-service solid waste providers and provide for their functions; the regulation, operation, maintenance and construction of landfills and other disposal facilities; and the setting and approval of tariffs for management of solid waste and provision of solid waste services.

Relevance

Much of the anticipated waste from the project is municipal solid waste in form of food waste, plastic and paper packaging during the operation period. Waste will be collected in designated receptacles and disposed of at the council licensed disposal site. The project site shall at all times be kept in a sanitary condition.

Compliance

Grand Prosperous will prepare a waste management plan for the sound management of the different waste streams that are likely to be generated from the campsites and all work areas.

Littering will not be allowed at operational sites, instead, receptacles for temporal storage of waste will be kept onsite. The garbage arising from the construction works and during operation of the

project, will be collected and stored in dedicated receptacles before being taken offsite for disposal at an approved disposal site.

As outlined in section 5 of the Act, solid waste shall be treated as a resource and Grand Prosperous shall promote a solid waste hierarchy for recovery of solid waste, where prevention, reduction, re-use, recycling and composting is of primary consideration followed by other means of recovery such as energy recovery and lastly disposal.

2.1.16 The Anti Gender-Based Violence Act, No. 1 of 2011

This is a Zambian law on gender-based violence which not only offers a comprehensive framework for protection, but also a means of survival for victims and survivors of gender-based violence, and prosecution of perpetrators. The Act specifically provides for:

- a) The establishment of a gender-based violence fund to assist victims and or survivors
- b) Establishment of shelter to support victims and or survivors of gender-based violence.
- c) Provision of emergency monetary relief
- d) Addressing of harmful traditional practices

The Act defines gender-based violence as "any physical, mental, social or economic abuse against a person because of that person's gender." In particular, the Act lists the following as amounting to gender-based violence:

- Violence that results in or is likely to cause physical, sexual or psychological harm or suffering to a person, including threats of such acts, coercion or arbitrary deprivation of liberty, whether occurring in public or private life; and
- Actual or threatened physical, mental, social or economic abuse that occurs in a domestic relationship.

The Act seeks to address all forms of gender-based violence. It goes on further to list the types of abuses arising from cultural practices to include forced virginity testing, forced marriages, "sexual cleansing" and child marriages. It also defines such abuse to include "abuse perpetrated on a person by virtue of the person's age, physical or mental incapability, disability or illness."

Relevance:

Mining exploration works can exacerbate risks of GBV as a result of interaction of workers with the local communities. The Act provides for mitigation and response to GBV.

Compliance

Measures to prevent the occurrence of gender-based violence will include ongoing sensitisation among the workforce about the need to avoid such anti-social behaviour and the consequences as provided in the law. The contractor will also ensure that victims have access to justice, assistance for telephone hotlines and women's resource centre that provide legal and psychosocial support to victims. Other measures will include the following:

- among the workforce, create public awareness of women's contribution to society and ensure women's input in decision-making.
- Create public awareness campaigns directed towards both men and women to enable greater
- awareness of their legal and human rights, the legal consequences of abusive behaviour, as well as the impact of GBV on future generations.
- Raise awareness of the importance of women's economic empowerment and the economic costs of the absence of women's contribution to the labour force as a result of violence.
- Initiate collaboration between community leaders, elders, local authorities and schools to create greater understanding of the link between GBV and poverty.
- Grand Prosperous to develop procedures to systematically share information on issues, legislation and policies connecting women's economic empowerment and men's burden sharing within the household.
- Support community participation and collaboration in forming social transformation forums including those committed to combating GBV.
- Establish partnerships with government bodies, NGOs, human rights groups, institutions, working on combating GBV.

2.1.17 The Pneumoconiosis Act No. 13 of 1994

This is an act to make new provision for the assessment and payment of compensation in connection with pneumoconiosis; to provide for the medical examination and standards of physical fitness to be required of persons exposed or likely to be exposed to the risk of pneumoconiosis.

Relevance

Mining related operations have the potential to expose workers to the risk of pneumoconiosis.

Compliance

All workers employed at the site will be required to show proof of physical fitness. Periodic certification will also need to be submitted.

2.1.18 The Ionising Radiation Protection Act, No. 16 of 2005

This piece of legislation was developed to establish the Radiation Protection Authority (RPA) and to provide for the protection of the public, workers and the environment from hazards arising from the use of devices or materials capable of producing ionising radiation.

Section 21. (1) of the act requires that any person who intends to import, process, mine, export, possess, transport, use, dispose of, or undertake any other activity relating to, radioactive material or any other source of harmful ionising radiation is should apply to the RPA.

Relevance

Some of the devices that may be used during the exploration works may cause radiation exposure by emitting ionising may emit ionizing radiation or releasing radioactive substances or materials.

Compliance

The RPA will be informed and permits obtained for the importation of any device emitting ionizing radiation. Workers handling such equipment will be protected by wearing appropriate clothing. Training will be also be provided.

2.1.19 The Chiefs Act, No. 13 of 1994

This Act to makes provision for the recognition, appointment and functions of Chiefs and Deputy Chiefs; for the exclusion of former Chiefs and Deputy Chiefs from specified areas in the interests of public order; and for the appointment and functions of chief returners.

According to section 3. (1) of the Act, only the president has the powers to recognize the office of the Paramount Chief, Senior Chief, Chief or Sub-Chief and until the President is satisfied that such person is entitled to hold the office under African customary law.

Sections 10 and 11 of the At spell out some functions of the chiefs which include preserving the public peace in his area and to take reasonable measures to quell any riot, affray or similar disorder which may occur in that area.

Relevance

The project area is located within the jurisdiction of a recognized chief who in this case is HRH Chieftainess Sikoongo.

Compliance

Grand Prosperous reached out to HRH Chieftainess Sikoongo before visiting the area and a consent letter (Appendix 1) was obtained. Grand Prosperous further pledges to continue holding

consultative meetings with the royal establishment and to provide updates on ongoing exploration activities.

2.1.20 The National HIV/AIDS/STI/TB Council Act, No. 10 of 2002

This Act establishes the National HIV/AIDS/STI/TB Council, defines its functions and provides for its composition; constitutes the Secretariat of the Council, and defines its functions and provides for its composition.

Section 4 of the act defines the functions of the Council which are to coordinate and support the development, monitoring and evaluation of the multi-sectoral national response for the prevention and combating of the spread of HIV, AIDS, STI and TB in order to reduce the personal, social and economic impacts of HIV, AIDS, STI and TB.

As read with the section 13 of the Act, the functions of the Secretariat include the following:

- a) ensure accessibility of HIV, AIDS, STI and TB information to the public throughout the country;
- b) assist in the development of guidelines for the screening of blood and blood related products in accordance with international standards in order to prevent the spread of HIV, AIDS, STI and TB;
- c) in consultation with health institutions and other stakeholders, develop guidelines for community based care for combating of HIV, AIDS, STI and TB and develop mechanisms for linkages between community based care and other health care systems.

Relevance

Where employees interact with the local people, taking into account improved earnings and coupled with unbecoming behaviour such as beer drinking, there is a possibility of the spread of HIV/AIDS.

Compliance

Grand Prosperous pledges to support any interventions by the National HIV/AIDS/STI/TB Council for the prevention and combating of the spread of HIV, AIDS, STI and TB. Sensitization campaigns on HIV/AIDS will be carried out among employees.

2.1.21 The Public Health (Infected Areas) (Coronavirus Disease 2019) Regulations, No. 22 of 2020

These regulations are developed in line with the provisions in sections 28, 30 and 114 of the Public Health Act with respect to areas infected by the Coronavirus Disease 2019 (COVID-19).

The regulations require that permission to enter or leave an infected area is granted by an authorized officer in a local authority. Regulation 7 gives powers to an authorized officer to enter premises to search for a case of COVID-19 or to enquire whether there is or has been a case of COVID-19. These regulations further demand that the nearest authorised officer in a local authority or a public health facility is immediately informed if one becomes aware or has reason to suspect that another person has died or is suffering from COVID-19.

Regulation 12 gives powers to an authorised officer to inspect public premises in order to ensure that it has sufficient sanitation and hygiene to prevent the occurrence or transmission of COVID-19. The authorised officer may order the cleaning or closure of a public premise or burial of any contaminated water body where the authorised officer determines that public premises or a water body does not have sufficient sanitation and hygiene to prevent the occurrence or transmission of COVID-19.

Relevance

The coming together of workers from different homes for operational activities, has potential to create grounds for transmission of the COVID-19. Improper personal health care among employees coupled with unsound sanitation facilities at the workplace can escalate the prevalence of the disease.

Compliance

Alongside awareness raising among employees, Grand Prosperous will ensure that the workplace has sound sanitation facilities that will be cleaned and disinfected on a regular basis. Handwashing facilities will be provided together with sanitizers. Social distancing will be practiced and employees will not be allowed to shake hands or unnecessarily mingle.

2.1.22 The Public Health (Notifiable Infectious Disease) (Declaration) Notice, No. 21 of 2020

These regulations were promulgated in accordance with the Minister of Health powers contained in sections 9 and 28 of the Public Health Act. The regulations relate to the declaration of the COVID-19 disease as a notifiable infectious disease.

Relevance

The coming together of workers from different homes work has potential to create grounds for transmission of the COVID-19. Improper personal health care among employees coupled with unsound sanitation facilities at the workplace can create grounds for the prevalence of the disease.

Compliance

In the event of any employee of employees contracting the virus, Grand Prosperous will ensure that the relevant authorities are informed and appropriate measures taken to prevent the escalation of the disease.

2.2 INSTITUTIONAL FRAMEWORK

Institutions relevant to the development of the proposed project include the following:

2.2.1 The Zambia Environmental Management Agency

The Environmental Management Act establishes ZEMA which is a statutory body within the Ministry of Water, Sanitation and Environmental Protection and the following are the main functions of ZEMA:

- Integrated environmental management and the protection and conservation of the environment and sustainable management and use of natural resources;
- The prevention and control of environmental pollution and environmental degradation;
- Provide for public participation in environmental decision making and access to environmental information
- Undertaking environmental auditing and monitoring; and
- Facilitating the implementation of international environmental agreements and conventions to which Zambia is a party.

In line with the EMA requirements, the mining exploration EPB will be submitted to ZEMA for approval after which ZEMA will undertake verifications and consultations with interested and affected (IAPs) prior to making a decision.

2.2.2 Ministry of Mines and Minerals Development

The Ministry of Mines and Minerals Development is responsible for policy formulation on matters pertaining to minerals development. The ministry is also responsible for coordinating and monitoring the other ministries that play a role in the effective conservation and management of the mining industry and the development of mineral resources. Since this proposed project is mining related, it falls under the jurisdiction of the ministry.

2.2.3 Chirundu Town Council

The District Council, headed by the District Council Chairperson and assisted by the District Council Secretary is the Local Authority, responsible for issues related to roads, sanitation, welfare and traditional authority (Chiefs). Councillors are chosen by the population to represent wards in the District Council. Full council meetings are held where all chosen councillors are present, with the local government as observers.

The District Development Coordinating Committee (DDCC), chaired by the DC and secretariat provided by the District Council, is the main body where all institutions, NGOs, faith-based organizations and other stakeholders in development come together to coordinate and inform the DC and subsequently the Provincial Development Coordinating Committee (PDCC).

2.3 INTERNATIONAL AGREEMENTS AND CONVENTIONS

Zambia is a signatory to a number of international Conventions. Most of the Conventions have been customized through various Regulations and are already being enforced. Applicable Conventions are briefly described below.

2.3.1 Convention on Biological Diversity (CBD) (ratified in 1993)

The objectives the Convention on Biological Diversity include, ensuring the conservation of a full range of natural ecosystems through a network of protected areas, development and implementation of strategies for conservation of biodiversity, sustainable use and management of biological resources.

Article 14 of the CBD part (a) states that: "Introduce appropriate procedures requiring environmental impact assessment of its proposed projects that are likely to have significant adverse effects on biological diversity with a view to avoiding or minimizing such effects and, where appropriate, allow for public participation in such procedures"

Relevance

There are a number of community forests which are managed by the communities through the local traditional leadership. Such forests are a habitat for diverse faunal species.

Compliance

Grand Prosperous will not allow unnecessary cutting down of trees in order to uphold sustainable environmental practices and in compliance with the provisions of this convention. Clearing activities will be limited to project operation sites.

Biological resources of significant conservation value that will be identified during Project implementation will be conserved and managed by Grand Prosperous with the involvement of the public.

2.3.2 Convention Concerning the Protection of World Heritage (1972)

The Convention aims at ensuring the identification, protection, conservation, presentation and transmission to future generations of the cultural and natural heritage.

Article 5 Part (a) states that "adopt a general policy which aims to give the cultural and natural heritage a function in the life of the community and to integrate the protection of that heritage into comprehensive planning programmes"

Cultural and natural heritage sites that may be identified during the proposed Project implementation will be protected and conserved in accordance with the provisions of the Convention to which Zambia is party to.

Relevance

The interaction of workers coming from outside Chirundu has potential to erode the cultural heritage of the local people. clearing and digging can lead to chance find of artefacts of cultural, historical or geological value.

Compliance

Grand Prosperous will ensure that the workers are obtained from the local community unless the required expertise cannot be obtained locally. A chance find procedure will be prepared to handle artefacts obtained through chance find.

2.3.3 The United Nations Framework Convention on Climate Change

The United Nations Framework Convention on Climate Change (UNFCCC) is an international environmental treaty also known as a multilateral environmental agreement that was opened for signature at the Earth Summit held in Rio de Janeiro in 1992 and came into force in 1994.

The ultimate objective of the Convention is to "stabilize greenhouse gas concentrations in the atmosphere at a level that will prevent dangerous human interference with the climate system. "It states that "such a level should be achieved within a time-frame sufficient to allow ecosystems to adapt naturally to climate change, to ensure that food production is not threatened, and to enable economic development to proceed in a sustainable manner." To-date 194 countries have since signed the UNFCCC showing near universal agreement that there is a problem and that action is required against climate change.

Relevance

The clearing activities have the potential to remove carbon sinks and thus contribute to localized climate change.

Compliance

Grand Prosperous will promote sustainable management of forests which will enhance the stabilization of greenhouse gas concentrations in the atmosphere at a level that will prevent dangerous human interference with the climate system.

3 PROJECT DESCRIPTION

3.1 PROJECT LOCATION AND ACCESS

The proposed project, exploration licence 36411-HQ-LEL, is located in HRH Chief Sikoongo's chieftdom in Lusitu area of Chirundu District. Access to the site is from the Siavonga (M15) Road at Lusitu, 5km before the Lusitu River, heading west for some 4km up to the eastern boundary of the project license (fig 1-1). Lusitu are is located some 16km south of the Chirundu/Siavonga (M15/T2) road junction.

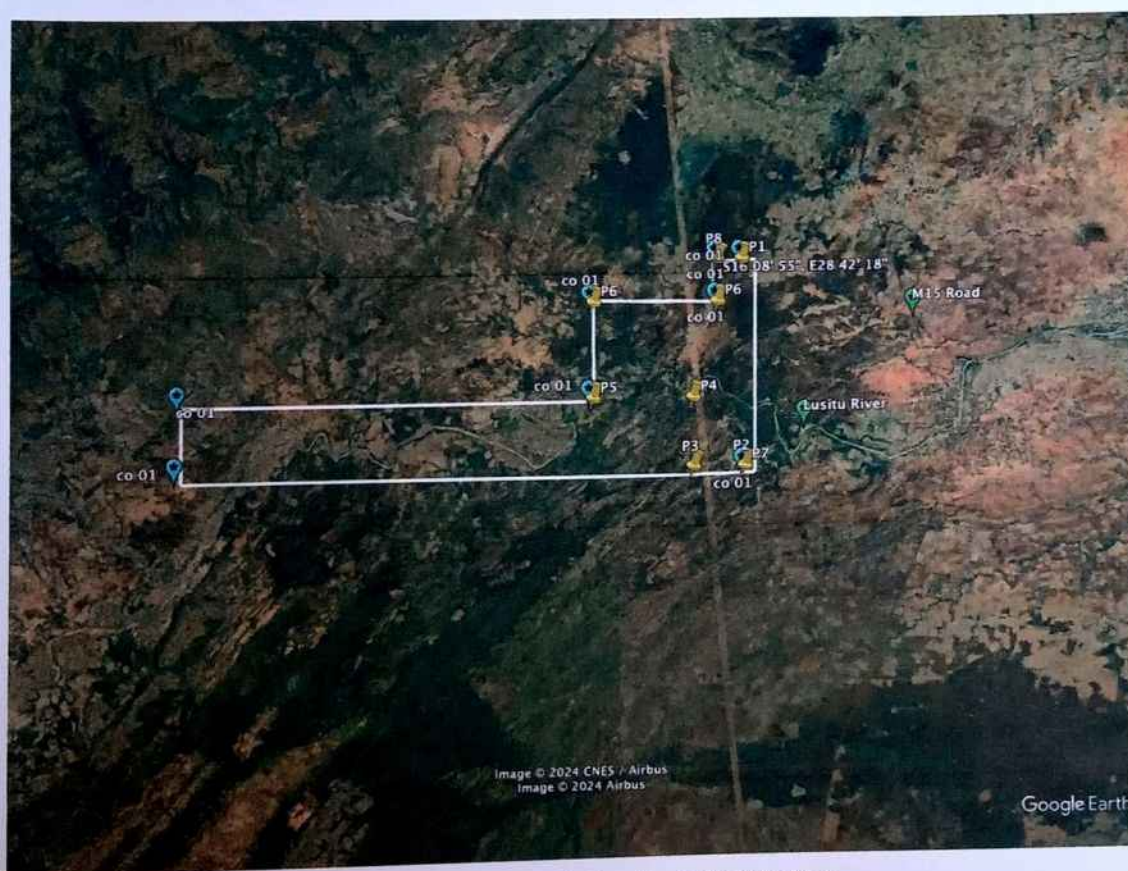


Figure 3-1. Google Earth showing project location

3.2 NOTABLE LANDMARKS IN THE AREA

The junction of the Chirundu and Siavonga roads is located some 16km north of the project site while the Lusitu village is 5km east of the project site. The Lusitu river meanders through the license area in a west to east direction. The M15 road is located 5km east of the project site.

3.3 NATURE OF THE PROJECT

The project is a mineral exploration project. Grand Prosperous will conduct exploration activities for cobalt, copper, iron ore, lead, nickel, silver and zinc. The project will involve the following mineral exploration activities:

- I. Site preparation: - some parts of the licence area will be cleared of vegetation. The areas that will be cleared will include access road, camping areas and drilling sites;
- II. Geological mapping: - Geologists will make ground traverses to map the geology of the licence area.
- III. Geochemical sampling: - This will involve sampling of geological materials such as soils, stream sediments, rocks and biological materials for geochemical analyses. These activities will be followed by plotting of the results on a map and interpretation of results;
- IV. Geophysical exploration: - this is the exploration of an area in which physical properties of rocks (e.g. resistivity, gravity, conductivity, magnetic properties) unique to the rocks in the area are quantitatively measured by one or more geophysical method. Equipment to be used is the resistometer, gravimeter and magnetometer
- V. Reverse Circulation Drilling (RCD): - This utilizes a solid bit to drill a hole and deliver rock chips to the surface. RCD activities are all automobile mounted hence the need for access roads; and
- VI. Diamond Drilling: - This method involves use of hollow drilling bits that are impregnated with diamond chips. As the bit rotates, it cuts a circular portion of the rock which is pushed up into the core barrel of the drill. The rock core is what is used for assaying.
- VII. Bulk sampling: - This will involve sampling of mineralized geological materials such as soils and rocks for mineral content analyses. These activities will be followed by plotting of the results on a map and generation of an orebody which will determine the appropriate mining method for the mineral resource.

3.3.1 Employment

Twenty-five (25) people will be employed during the prospecting works.

3.3.2 Schedule and Lifespan of the Project

The project is expected to commence as soon as all the approval requirements are met and done through the various relevant line organizations and institutions which include the EPB approval process by ZEMA. It will take a span of three years to be completed.

The timeline of the project is presented in table 3-1 presented below.

Table 3-1. Timeline for project implementation

No.	Task Name	Duration	Start	Finish
1.	Preparation	3 months	May 2024	Jul 2024
2.	Operation	36 months	Aug 2024	Jul 2027
3.	Decommissioning and Closure	3 months	Jul 2027	Sep 2027

3.3.3 Raw Materials

Different types of raw materials will be used during exploration and these have been tabulated in Table 3-2 which include diesel, lubricants, mobile equipment consumables (batteries, filters, tires etc), mobile machinery and vehicles, food for workers (canteen), water, timber, and camp construction material.

Table 3-2. Project Inputs, Outputs and by products

Input	Product	By product/waste
Drilling	Geological samples	Drill fluids; soil
Diesel	Powering machines and service vehicles.	Gas emissions
Lubricants	Smooth operations of machines	Hazardous waste
Mobile equipment consumables (batteries, filters, tires etc)	Smooth operation of machines	Hazardous waste
Mobile machinery and vehicles	Exploration activities	• Vibration • Hazardous waste • Scrap metal
Food for workers (Canteen)	Nutrition supplement to workers	• Domestic waste generation (canteen garbage) • Sewage waste
Water	Dust suppression	Waste water
Timber	Wooden structures	Timber off-cuts
Camp construction material	Accommodation for workers	• Scrap metal • Nails • Tree off cuts • Wood off cuts • Empty paint cans

The diesel for running the machines will be stored on site in dedicated 210 litre drums. Only 4 drums will be kept onsite at any given time. All the equipment will be taken to the workshops in Choma or Monze, when they are due for service.

Fieldwork will be conducted from base camps that will be set up in suitable locations in the licence area. The camp sites will be located at least 300 metres from the nearest surface water body. Structures at these base camps are usually environmentally friendly and high standard of

cleanliness will always be adhered to. Bushfires shall always be avoided to ensure that flora, fauna and water resources are protected.

Existing access roads will be rehabilitated and new access roads will be constructed where necessary.

Camp sites will comprise of a maximum of ten tents and two (2) portable toilets. A campsite will accommodate a maximum of 15 people. Some tents will accommodate two (2) people. The rest of the people that will not be accommodated at the camp will be coming from their homes.

The tents will be constructed using the following materials;

- Canvas sheets
- Metal poles and pegs
- Ropes

Camp site construction will generate waste like;

- Food remains
- Tree off cuts
- Nails

Prudent waste management shall be applied (re-use of materials such as trees cut from site clearing will be used as fire wood, etc). The main source of energy will be gas stoves. Waste bins shall be provided for the kitchen and offices and these shall always be provided with bin-liners to avoid double handling. Littering shall not be acceptable and reminder signs shall be installed to inform workers and visitors that all waste must be thrown in designated waste bins. Waste disposal shall be per ZEMA recommended standard (only registered refuse collectors shall be used). The waste shall be disposed of at a ZEMA licensed dump in Chirundu. Records of the waste disposed of per week shall be maintained.

Portable toilets will be used for sanitation facilities. The portable toilets will be disinfected using lime. Such facilities shall be located at least 100m away from any water body.

No firewood will be used for cooking. Drinking water at the camp will be drawn from the local borehole.

cleanliness will always be adhered to. Bushfires shall always be avoided to ensure that flora, fauna and water resources are protected.

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No firewood will be used for cooking. Drinking water at the camp will be drawn from the local borehole.

3.3.4 Day-to-Day Waste Management and Pollution Control Activities

Day-to-day waste management and pollution control activities will include the following procedures;

- Waste separation at the point of generation shall be a fundamental activity and appropriate signs shall be painted on waste bins to facilitate separation.
- Biodegradable waste shall be collected in waste bins and dumped at the nearest ZEMA licensed dumping area.
- Domestic waste bins shall be distributed at appropriate places and these shall be labelled for waste disposal. Bin-liners shall always be provided to avoid double handling.
- A "Please No Littering, Use Waste Bins" sign shall be installed at the gate to remind everyone entering the project premises that littering is not acceptable.
- Weekly records of biodegradable waste bins collected shall be kept by the head of housekeeping and these shall be reported in monthly reports.
- All the salvageable material shall be kept within a salvage yard which shall be created within the camp. The material that shall be kept in the salvage yard shall include used tires, scrap metal and any material that can be re-used or sold.
- Records of the commercial waste and domestic waste generated per month shall be kept by the Project Manager and these shall be reported in monthly reports.

3.4 MAIN ACTIVITIES

The project has three main phases; Preparation Phase, Operation Phase and Decommission and Closure Phase.

3.4.1 Preparation Phase

Major activities shall include recruitment of workers, clearing access routes, and community liaison.

Other activities include the following:

- Major activities will include construction of temporary camps and sanitary facilities.
- Since the exploration site is in undisturbed area, no people will be relocated.

3.4.2 Operation Phase

The main exploration activities shall include the following:

- Geological Mapping: - This will involve geologists traversing the area to map the rock formations of the project area.
- Geochemical sampling: - This will involve sampling of geological materials such as soils, stream sediments, rocks and biological materials for geochemical analyses. These activities will be followed by plotting of the results on a map and interpretation of results;
- Pitting and Trenching: - Pits (20cm x 20cm dug to depths of 3-4m) and trenches will be dug to expose shallow sub-surface geology. The pits will be buried as soon as sampling and logging is completed. Generally, excavated depths of up to 4m for the trenches will be dug.

- Clearing of Drill Sites: Drill sites will be cleared in preparation for the drilling activities.
- Reverse Circulation Drilling (RCD): - This utilizes a solid bit to drill a hole and deliver rock chips to the surface. The rock chips are used for assaying. RCD activities are all automobile mounted hence the need for access roads; and
- Diamond Drilling: - This method involves use of hollow drilling bits that are impregnated with diamond chips. As the bit rotates, it cuts a circular portion of the rock which is pushed up into the core barrel of the drill. The rock core is what is used for assaying.
- Bulk sampling: - This will involve sampling of mineralized geological materials such as soils and rocks for mineral content analyses. These activities will be followed by plotting of the results on a map and generation of an orebody which will determine the appropriate mining method for the mineral resource.

3.4.3 Decommissioning and Closure Phase

The main activities that shall be undertaken under this phase shall include;

- Demolition of temporary houses and sanitary facilities
- Sealing of drill holes
- Burying of exploration trenches and pits
- Posting warning signs
- Removal, treatment and sale the of 210 litre fuel drums
- Demolition of concrete infrastructure and levelling the surface
- Treatment of hydrocarbon contaminated portions covering the project site
- General site clean-up

*handover to the community/headsman
learn to build permanent structures*

4 ANALYSIS OF THE PROJECT ALTERNATIVES

Table 4-1. Project alternatives

Alternatives	Preferred alternative	Reasons
• Soil sampling • sediment sampling • trenching and • drilling	All these alternatives shall be used	Combination of exploration methods are the best methods of conducting exploration activities
Pitting and excavation • Hand-held picks and shovels vs • the use of an excavator	Hand-held picks and shovels	Hand-held picks and shovels were preferred because they make smaller excavations and hence are environmentally friendly as compared to the excavator
Drilling methods • Reverse circulation drilling (RCD) • diamond drilling (DD)	Both types of drilling will be employed	Diamond core drilling is good for precise sampling and analysis of layers while RC is good at extracting bulk samples.
Site alternatives	Alternative site could not be considered because the licence is site specific.	The choice of where to conduct exploration is largely determined by availability and issuance of licences by the Ministry of Mines and Minerals Development. Project operation sites for further mineral prospecting are determined by the potential presence of anomalies.
Fuel storage alternatives • The use of a 5000-litre fuel bowser • The use of dedicated fuel drums located at the campsite	The use of a 210 litre fuel drums was preferred because of cost implications	Considering the temporal nature of the project, the use of a 5000 litre tank is not practical and is costly. Drums are easy to relocate and are cheap.
Power supply alternatives • Diesel generators • Solar panels • connecting to the grid • battery • gas stoves	Solar and battery for lighting Gas stoves for cooking	Solar panels were preferred because they are easier to transport and install and are relatively environmentally friendly. There is no ZESCO grid in within or near the site Battery powered sources will be used for lighting. Generators will require the use of fuel. Coal has to be transported from Maamba and the costs involved are prohibitive while the firewood is not sustainable.
Water sources • Sinking own borehole • Community well or borehole • Surface water	Sinking own borehole (or drawing water from the community well or borehole where these are available)	Absence of surface water in some areas of the project area. However, for some operational areas which are near perennial streams, surface water will be obtained.
Municipal waste disposal • Having own disposal site • Using the council owned waste disposal site • reusing some waste streams • burying and burning	Using the council owned waste disposal site Reusing some waste streams	Having own disposal site has the potential to cause aesthetic impacts and has operational challenges Burying can pollute land and water resources Burning of waste pollutes the air and is offensive to neighbours Biodegradable waste such as putrescible will be buried for the generation of manure while reusable items such as packaging waste will be given to the local people
Night soil disposal • connecting to the provincial sewerage system,	Portable toilets	Connecting to the sewer system was dropped because of the costs involved and the temporal nature of the camp sites. Drop-down

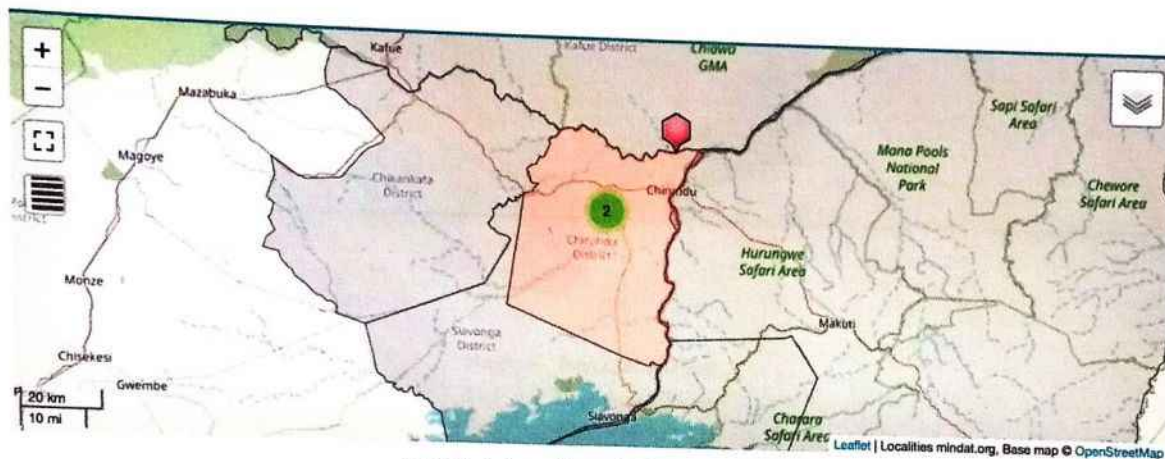
• using onsite drop-down toilets • portable toilets that can be offloaded periodically		toilets are also not preferable due to the potential to contaminate groundwater. Since the construction camps will be temporal, portable toilets will be used at camp sites.
• No project option • Going ahead with the project	Going ahead with the project	The "No Project Option" would negatively affect the development of the mineral resource and hence deprive the country of the much-needed foreign exchange that the project would earn through the export of minerals if the exploration results into mining. This option would also deprive the local people of the much-needed employment opportunity and improved local businesses.

5 BASELINE ENVIRONMENTAL AND SOCIAL CONDITIONS

5.1 GENERAL

Before 2012, Chirundu was part of Siavonga District in the Southern province of Zambia. President Michael Sata decided in 2012 to create the Chirundu District (which consequentially led to the Chirundu Constituency being created for the next general elections), effectively declaring that Chirundu will have its own town council administration (no-longer under Siavonga).

Chirundu is bordered by Kafue District to the north, Chikankata District to the west, Siavonga District to the south and Zimbabwe to the east (Map 5-1).



Map 5-1. Location of Chirundu District

Source – www.mindat.org

Chirundu district has the following six wards – Chirundu Ward, Ibwe munyama Ward, Lusitu Ward, Musaya Ward, Ng'ombe Ilede Ward and Sikoongo Ward.

5.2 PHYSICAL ENVIRONMENT

5.2.1 Climate

5.2.1.1 Temperature

The daily mean temperatures may vary from 20°C to 25°C. During summer, actual temperatures go as high as 40 Degrees Celsius. Temperatures reach a maximum value from October to December, just before and following the onset of the rainy season. In the rainy season (from November to March), temperatures gradually decrease, and after the rainy season, this rate of decrease becomes large and temperatures fall until June-July.

5.2.1.2 Humidity

Relative humidity shows minimum values of around 30-40% from September to October, and then rises rapidly with the beginning of the rainy season. After December, humidity is maintained at 60-80% until April, and then gradually decreases.

5.2.1.3 Rainfall

The project area is located within agro-ecological Region II (Fig 5-1) and receives an annual rainfall of less than 700mm falling between November and mid-April. Rainfall is unpredictable, and in some years, there is hardly any, making the valley a drought prone area.



Region	Annual Rainfall
I	Less than 800 mm
IIa	800 - 1000 mm
IIb	800 - 1000 mm
III	More than 1000

Figure 5-1. Agro-ecological regions of Zambia
Source: *Zambia Livelihood Zones and Descriptions* (FEWS, 2014)

5.2.2 Air Quality

The air quality is good in the area and the exploration area is in an undisturbed forested area. Therefore, there is no air quality data available for the proposed project area. Field observations indicate that the general air quality in the area is good. However, seasonal variation as well as localized and temporal deterioration in air quality does occur.

Particulate matter was analysed using the gravimetric method by sampling with small, yet effective, filter samplers equipped with PM₁₀ and PM_{2.5} cut size inlets. The passive sampler collects particles by gravity, diffusion, and convective diffusion onto a glass coverslip that is then examined with an optical microscope; digital images are processed with software and the resultant PM_{10-2.5} concentrations determined. Sampling was done for periods of two days. Particles consisted primarily of inorganic dusts. Measured concentrations ranged from < 11 microg/m³ to approximately 15 microg/m³. Daily averages of PM_{10-2.5} were 8.6 µg m⁻³.

5.2.3 Noise Levels

The noise levels recorded at the project site are as shown in Table 5-1.

Table 5-1: Average noise level around and within the project area

Location/Description	Noise reading (dB)
Lusitu Village	55.4
Lusitu River	55.2
Lusitu Primary School	46.7

The project area is located in a rural setting, in Lusitu village, where there are isolated settlements or developments. The sources of noise within the area are from natural sources such as different weather conditions wind, rain and thunderstorms. During natural conditions noise levels varies generally between 20 dBA and 50 dBA but can occasionally increase significantly during heavy rains and thunderstorms.

5.2.4 Geology

Zambia has complex geographical features embracing both basin and plateau characteristics owing to differential movement and buttressing effects of cratons sand-witching the country (Fig 5-2). Further, its complex geology coupled with a multiplicity of tectono-thermal events have shaped the obtaining geological makeup of the country. The final in the series of these tectono-thermal events was the Karoo Rifting associated with the break-up of Gondwanaland during the Permian followed by opening of the proto-Indian Ocean in the Jurassic; and a final episode of rifting related to the development of the East African Rift system of which the Zambezi Rift Valley is in the late Cretaceous and early Tertiary times.

The middle Zambezi Rift Valley in which the project site is located is thus a trough formed by down-faulting dating back to the Karoo period and as such is underlain by rocks of the Karoo Supergroup. The Karoo Supergroup (Table 5-2) is subdivided into Upper and Lower Karoo. The Lower Karoo Supergroup mainly comprise of fine clastics while that of Upper Karoo consist of coarse clastics.

The Lower Karoo Group comprises a basal conglomerate, tillite and sandstone overlain unconformably by conglomerate, coal, sandstone and carbonaceous siltstones and mudstones (the Gwembe Formation), and finally fine-grained lacustrine sediments - the Madumabisa Formation. The unconformably overlying Upper Karoo essentially comprises a series of arenaceous continental sediments and overlying mudstones capped by basalts of the Batoka Formation.

The occurrence of coal in the area is associated with Lower Karoo rock formation while Upper Karoo is characterised by Uranium mineralisation. The project site is thus underlain by Lower Karoo rocks with sandstone, mudstone and coal formations.

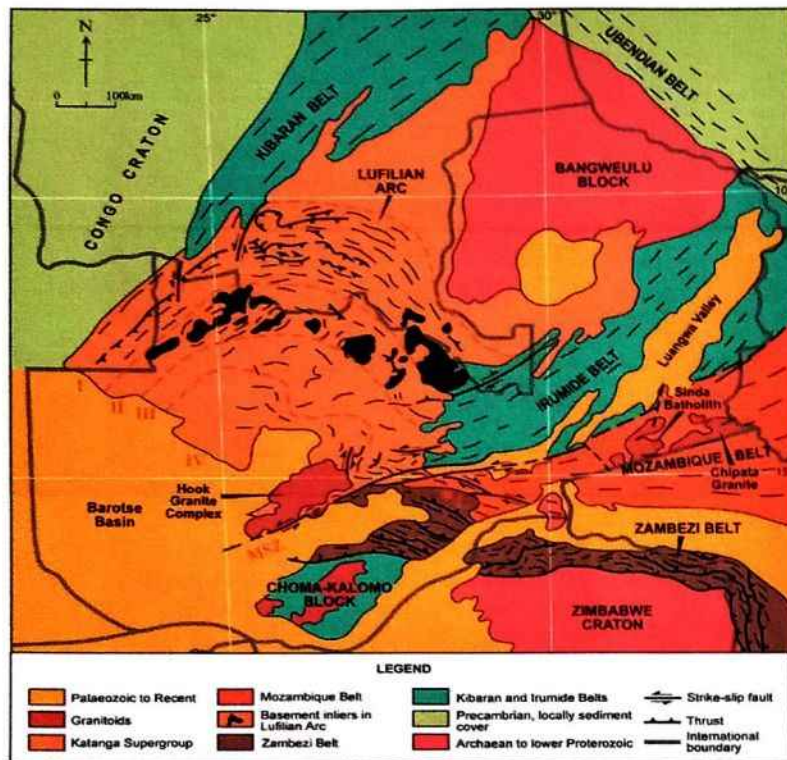


Figure 5-2. Zambia - Geological Terrain

SUPERGROUP	MAJOR GROUP (and some equivalents)	OROGENY	
Karoo	Katohani	KAROO RIFTING	
	Lower Formation		unconformity
	Upper Karoo		
	Lower Karoo		unconformity
	Katiba		unconformity
Kuberga	Kandabunga	unconformity LUFILIAN OROGENY (D2 & D3) c. 690-450 Ma	
	Mweenche Fm. (Mweenche Formation)	unconformity tectonic LUFILIAN OROGENY (D1) c. 850-750 Ma	
	Miba Series (Broken Hill) (Morros) (Soweto)	Lusakan Folding Event c. 850 Ma LOMANIAN OROGENY c. 950 Ma unconformity	
Mava	Manahya River	unconformity IRUMIDE OROGENY c. 1350-1100 Ma	
	Mpanahya (Tosare) (Mitoba River)		
	Mpanokoso (Kafun) (Fombwe)	tectonic & unconformity	
Boswanant	Kapemba (Nakonde)	tectonic USENZIAN OROGENY c. 2000-1800 Ma	
	Mukungud (Muyge) (Mkushi)	tectonic	

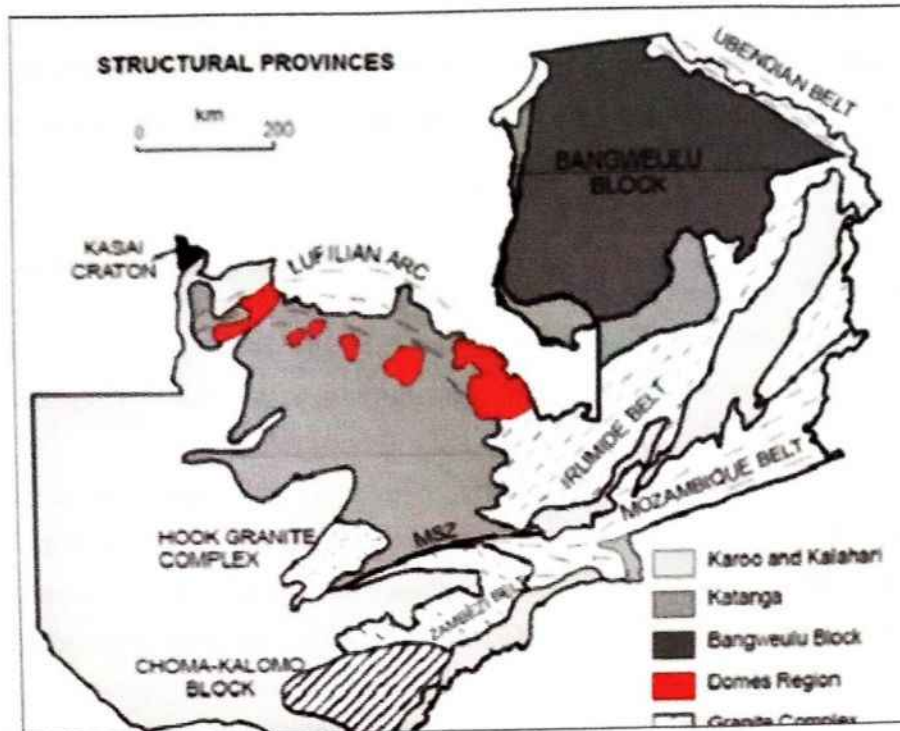


Figure 5-3. Zambian geology

5.2.5 Soils

As noted from the geology discussion above, the rocks of the valley floors date from the Karroo period giving rise to soils mostly derived from the Karroo sandstones and mud stones. These soils generally have a higher mineral content than those derived from the basement complex of the plateaux.

Soil characteristics in the area vary following the area's transition from upper ground on the Zambezi Escarpment down to the Zambezi Valley. Soils are slightly shallow on hills and escarpments which are characterised by rocky outcrops with improvements occurring down slope as the soil profile develops. Valley bottoms have much well-developed soils which are a combination of sandy and clay soils with deep profiles.

Dambo areas are mostly dominated by clayey soils most of which are poorly drained and have poor aeration. These clays swell on absorbing water, and rapidly become impervious but crack when dry thereby opening up the soil.

On dambo fringes are isolated termite mounds which are characterised by soils rich in mineral salts representing islands of rich soils even where they are surrounded by mineral deficient soils.

Most parts of the district have poorly developed soils mostly sandy and this gives rise to soil erosion with widespread gully formations.

Like the rest of the district, soils in the project area are sandy with isolated patches of sandy clays and clayey sands. Soil coloration ranges from light to dark with clayey soils manifesting darker coloration.

5.2.6 Landscape and Topography

The district is characterised by outstanding geomorphologic features, which are spatially interrupted by hills with lofty summits. On the northern and north-eastern sides is the Zambezi escarpment, which makes most parts of the district inaccessible in terms of road transport.

The terrain is mountainous with steep slopes, characterized by fast flowing, fast drying ravines undulating and cascading rocky terrain which result in the land being highly erosive, leaving behind gullies. Lusitu is situated at an average elevation of about 1350m above sea level. The area forms a gently undulating plateau rising to low hills of about 600m dropping area to about 420m. The area mainly drains towards the south into Lake Kariba. Lake Kariba is the main and ultimate drainage of the area and its surrounding.

5.2.7 Hydrology

The main water course draining the project area is the Zambezi River located to the east of Lusitu. There are also other streams which are tributaries of the Lusitu River draining into the Zambezi River. These streams and rivers are dry for most of the year due to the steep gradients, which give rise to high overland flow and drainage into Zambezi River.

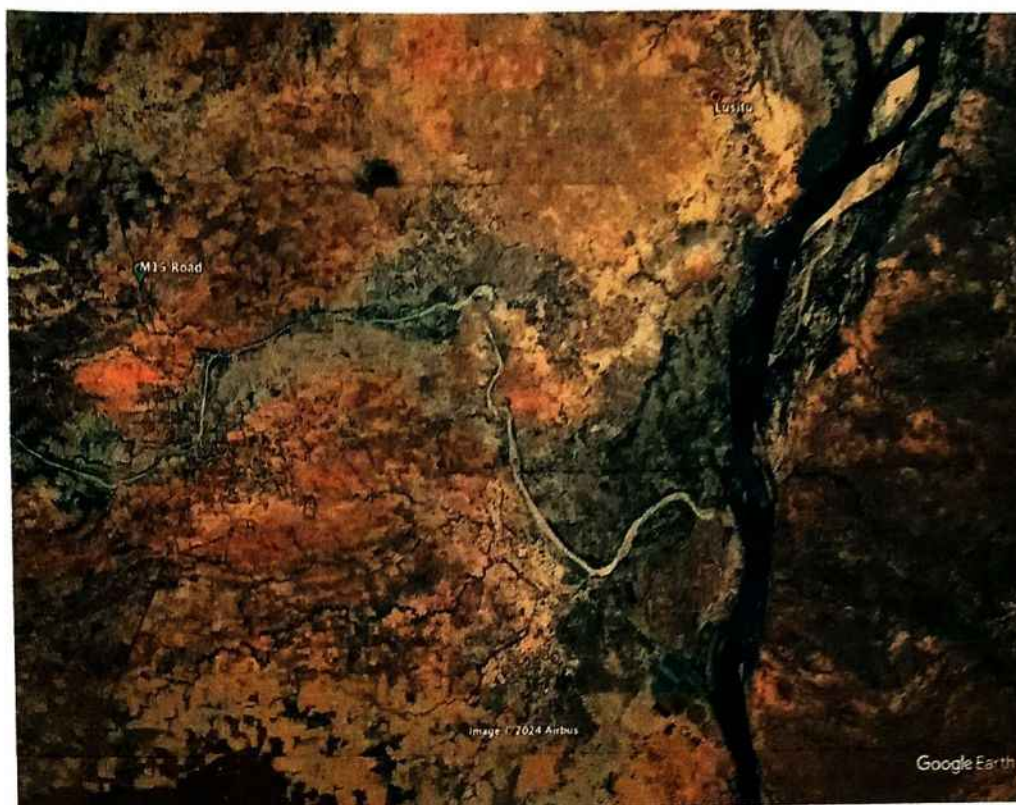


Figure 5-4. Drainage pattern of the area

Some rivers and streams drain from the Zambezi escarpment making some areas impassable during the rainy season. The streams, when filled due to flush floods, may also wash away some bridges which are not strongly built. Along the sides of these rivers and streams are rich flood plains of alluvial soils, which are cultivated by the local people. The upper valley joins the escarpment while the larger part of lower valley has been buried beneath Lake Kariba. There is a lot of crop cultivation being done on the escarpment due to relief rain and people have settled along the escarpment.

The project is characterised with steep gradients giving rise to streams that traverse the area and flowing into Lake Kariba. These streams were dry at the time of the survey in July 2024.

5.3 BIOLOGICAL ENVIRONMENT

5.3.1 Flora

Owing to its location in the Zambezi valley which forms part of the Rift Valley, vegetation in the Lusitu area is typical rift valley savannah with a mix of deciduous woodlands, scrubland, thickets, riverine and patches of grassland with variation from site to site as dictated by the underlying rock, soil type, drainage and altitude.

According to Trupnel (Vegetation classification of Zambia, 1969), Lusitu vegetation falls under Mopane vegetation, largely with *Colophospermaum mopane* as the dominant tree species. However, field surveys conducted in the area indicated that the area has mixed vegetation that included; riparian, open woodland with mixed tree species. The amorphous arrangement of settlements with agriculture practices coupled with poor rainfall patterns (average annual rainfall range of 625 mm – 677 mm), has led to fragmented and stunted vegetation in the area to such an extent that most plant species are at shrub level.. Therefore the proposed project will not have any significant impact on flora in the area because the area is already depleted of vegetation mainly due to clearing of land for settlements and crop cultivation.

The Acacias such as *faiderbia albida* trees are common species found mostly around the farm with isolated *piliostigma thonningii* tree species seen within the area. The surrounding characteristic natural vegetation is dominated by some Acacias and *Sparthodia spp* trees (mulele) often used to get fibre for thatching village huts and *Hyperhenea* (the thatching grass) among the many other plant species. Small shrubs were also observed around the project area.

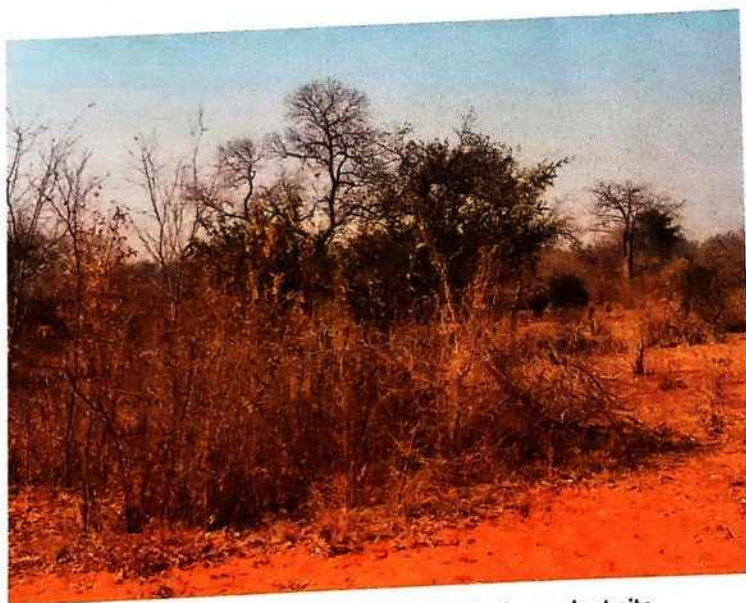


Photo 5-1. Typical vegetation within the project site



Photo 5-2. Baobab trees are common within the project area

5.3.2 Fauna

There were no signs of either large or small wild animals (game) on the site. According to the locals, the following have been cited:

Table 5-3. Large mammals of the project area

Class/ Order/Family	Latin name	Common name
Order Carnivora		
Canidae	<i>Canis adustus</i>	Side-striped jackal
Felidae	<i>Felis sylvestris</i>	African wild cat
	<i>F. caracal</i>	Caracal
	<i>F. serval</i>	Serval
Order Proboscidea	<i>Loxodonta africana</i>	Elephant
Order Artiodactyla		
Suidae	<i>Potamochoerus porcus</i>	Bushpig
	<i>Phacochoerus aethiopicus</i>	Warthog
	<i>Hippopotamus amphibius</i>	Hippopotamus

Other species include the following given below.

Table 5-4: Amphibians

Scientific Name	Common Name
<i>Breviceps poweri</i>	Rain frog
<i>Bufo lemairi</i>	Yellow toad

<i>Bufo regularis</i>	Camouflage toad
<i>Rana occipitalis</i>	Giant swamp frog

Table 5-5: Reptiles

Scientific Name	Common Name
<i>Bitis arietans</i>	Puff adder
<i>Dendroaspis polylepis</i>	Black Mamba
<i>Geochelone species</i>	Tortoise (various)
<i>Naja naja</i>	Common Cobra
<i>Naja nigricollis</i>	Spitting Cobra
<i>Python sebae</i>	Python
<i>Varanus niloticus</i>	Water Monitor

Table 5-6: Insects

Scientific Name	Common Name
<i>Anisoptera spp</i>	Dragon fly
<i>Pantodon buchholzi</i>	African Butterfly
<i>Ophion spp</i>	Wasps
<i>Formicidae spp</i>	Red Ants
<i>Apis mellifera</i>	Bees
<i>Anopheles quadrimaculatus</i>	Malaria transmitting mosquitoes

The project site is essentially cleared of fauna habitat and has minimal value in this respect. There are no known rare or endangered animal species in the project area.

5.3.3 Birds

Some of the local birds include; *Pycnonotus barbatus* (Pwele-Bemba), *Acrocephalus baeticatus* (titi - Bemba), Common Bulbul (*Pycnonotus barbatus*), African Yellow White Eye (*Zosterops senegalensis*), Black Collared Barbet (*Lybius torquatus*), White Bellied Subbird (*Cinnyris talatala*), Amethyst Sunbird (*Chalcomitra amethystina*), Starling (*Cinnyricinclus leucogaster*), Yellow Fronted Canary (*Serinus mozambicus*) Others included, the Robin chats (*Cossypha semirufa*), Marico Flycatcher (*Bradornis mariquensis*), Common Fiscal (*Lanius collaris*), Bronze Mannikin (*Lonchura cucullata*), Scarlet Chested Sunbird (*Chalcomitra senegalensis*), Green Backed Honey Guide (*Prodotiscus zambesiae*) and the African Yellow Bellied Green Bull (*Chlorocichla flaviventris*).

Other bird species included; black cheeked love bird (*Agapornis nigrigenis*), finfoot (*podica senegalensis*), spoonbill (*Platelea lba*), long-crested eagle (*Lophaetus occipitalis*), western banded snake eagle (*circaetus cineroscens*), and African skimmer (*Rynchops flavirostris*). The rare or

endangered bird species likely to occur within the project area included, Helmeted guinea fowl (*Numida meleagris*) Hornbill (*Tockus spp*) and rock pratincole (*Glareola nuchalis*).

Common water birds include spurwing goose (*Plectropterus gambensis*), crowned crane (*Bulirica Regulorum*) Openbill (*Amustomus Lamelligerus*), Francolin (*Francolinus swainsonii*), Saddle billed stock (*Ephippiorhynchus senegalensis*) and a variety of duck species. Due to the timing of the survey, most migratory birds and other local resident species could have not been recorded.

Nocturnal birds such as the owl were common around the forest areas.

5.4 SOCIO-ECONOMIC ENVIRONMENT

5.4.1 Human settlement

The population distribution follows the physical characteristics of the district. Lusitu is located in a rural area. The settlement pattern tends to be influenced by proximity to sources of water and easy access to road network. The population is thus concentrated along the M15 and near the Lusitu river, which is unfortunately dry at the time of the site visit. The hilly nature of the general project affects settlement as families tend to settle around flat areas as opposed to hilly terrain.

5.4.2 Administration

The locals have representation in the National Assembly elected Members of Parliament, whose term of office is five years. At the district level, the highest ranking government official is the District Commissioner who oversees all the operations of government institutions in the district.

Each ward has an elected Councillor. The group of councillors form a council – the supreme body of the local authority, which plans and approves developmental programs and plans in the district. This council has an administrative wing which is headed by the Council Secretary.

Traditionally, each village is headed by village headmen who are also under the rulership of the chief.

5.4.3 Traditional Authority and Ethnic Groups

Traditionally, the project area is under the leadership of HRH Chieftainess Sikoongo, with the prominent ethnic group being the Valley Tonga.

5.4.4 Local Authority

Lusitu area is under the jurisdiction of Chirundu Town Council.

5.4.5 Population

The 2022 Census of population and housing indicated that Chirundu had a total population of 78,780 disaggregated in terms of sex as 38,215 for males and 40,565 for females. The population grew from 47,344 in 2010 representing an average annual growth rate over the 2010-2022 intercensal period of 4.3 percent. Having a land area of 1,386.0 square kilometers, Chirundu District has a population density of 56.8 persons per square kilometer. Source: ZamStats 2022 Census of population and housing report

Further information about the population structure for Chirundu shows that there are 43,583 people living in rural areas while 35403 people live in urban areas.

5.4.6 Socio-economics activities

5.4.6.1 Agriculture

Agriculture remains the major economic stay of the people in the area. Small holder farmers of Lusitu have been able to produce okra, tomatoes and bananas heavily dependent on rain-fed agriculture supplying to markets of Siavonga, Chirundu Kafue and sometimes Lusaka. With the Lusitu irrigation scheme being developed with the support of the World Bank, farmers will now be able to grow crops through the year in this valley of a drought-prone area with harsh desert-like climatic conditions.

Other sources of income include charcoal burning and roadside trading which deals in products such as handcraft, groceries and agricultural produce.

5.4.6.2 Trading

A number of roadside retail shops and tuntembas were observed, especially near the main and access roads. The shops trade in groceries which are obtained from Kafue, Chirundu or Lusaka. Road side trading in charcoal and agricultural input supplies and produce, especially in vegetables was also observed.

Trade is mostly in form of agricultural produce such as maize, chickens, soya, and goats which are sold within major market centres in Kafue or Lusaka while the M15 and T2 roads facilitate movement of goods within and outside the project area. Households tend to sell their produce to private traders, who, in turn, sell them on major markets in Lusaka and the Copperbelt.

The selling of agricultural produce provides cash for the locals to pay for children's school fees and buy products such as soap, clothing, cooking oil, sugar and other groceries are imported.

Livestock production in form of cattle, goat, chickens and pigs is widely practiced in the area.

5.4.7 Infrastructure

5.4.7.1 Telecommunication

Lusitu has fairly well-established infrastructure of domestic and international telecommunications, which is controlled by the Zamtel. Cellular phone services are also now available via Zamtel and a number of other privately owned companies such as MTN and Airtel.

Celullar service at the project site is available within the license but poor in some areas owing to the uneven terrain.

5.4.7.2 Energy

The project area is. Not connected to the national electric grid. As a result, the people in rural areas turn to alternative sources of energy which are unsustainable and non-renewable. The primary sources of energy in are, biofuels (firewood and charcoal), solar, and battery powered. For many years firewood and charcoal have been the main cooking fuel due to financial limitations.

5.4.7.3 Roads

Road infrastructure in the project area is bad to fairly good. One major bitumen road (M15), that connects Siavonga and Chirundu, passes through Lusitu enroute to Siavonga. The rest of the roads leading to other areas of the district are gravel roads most of them in a bad state.

5.4.7.4 Water and sewerage

Due to the lack of sustainable surface water, the local people obtain the water for domestic use from boreholes that have been sunk with the support from the government and non-governmental organisations. The streams in the area are seasonal and run dry during the drier months of the year.

CTC in collaboration with the Zambezi River Authority (ZRA) will invest a sum of more than K3 million towards rehabilitating the Lusitu Water Reticulation System. In this project, CTC signed an agreement with the Southern Water and Sewerage Company to handle the project. The Project covers 5 wards namely Njame, Mushungu, Nkadabbwe, Ng'ombe Ilede and Lusitu. The named areas have been grappling with the issue of salty water for many years but the Council in conjunction with other stakeholders will address the challenge.

Chirundu Town Council has also completed drilling and equipping of solar powered boreholes under CDF for the year 2023.

At traditional homesteads, drop-down toilets are a common feature.

5.4.7.5 Health

Health care services are provided at the Lusitu Rural Health Centre.

5.4.7.6 Education Facilities

Lusitu Primary school provides primary-level education while secondary education is provided at the Lusitu Secondary school.

5.4.7.7 Housing

In the village set up, the buildings are made of wattle-and-daub, and circular in plan, although there are a few rectangular huts. The verandah is enclosed round three sides of the hut. This enclosure forms an extra room which can be used either as pen for small stock or as a kitchen. Houses of sun-dried brick are now being built. The huts are laid out in rows, with the doors facing one another form rudimentary streets. The grain-bins, kitchens huts, and latrines are sited behind the living huts. The grain-bins are of two kinds, one much larger than the other. They are cylindrical in shape, and raised from the ground on platforms.

5.4.8 Water and Sanitation

There are limited sources of water available to provide clean drinking water to the local population. Surface water sources are often highly polluted, and infrastructure to pipe water from fresh, clean sources to arid areas is costly.

Most rural water supplies in the area are community-based. That is to say, most water systems, in this case boreholes, hand pumps and mechanised water systems, are owned, operated and managed by a community rather than an individual or household.

5.4.9 Land-use

The project area, Lusitu, is deemed a rural area and has vast undeveloped land which is covered by trees. Much of the area is covered by mopane woodland dominated by acacia. there is evidence of deforestation due to clearing of land for farming, grazing and charcoal burning. This deforestation has in turn probably resulted in loss of biodiversity, though this has not yet been quantified. The hot climatic conditions do not make it easy for the forest areas to be naturally replenished by the rains.

There are patches of settlements dotted especially along the main road, M15.

The main land use activities in the project area include pastoral and crop farming.

5.4.10 Archaeology

The Chirundu Petrified Forest is an area of fossil trees 21 km west of Chirundu, lying just south of the Chirundu - Lusaka Road near the Kariba turn-off. It is a listed national monument of Zambia.

Ngombe Ilede is another national monument located 7km east of the project site.

The local people do not have permanent community grave yards but use back yards within their homesteads. Consultations will need to be conducted with the local people.

The digging and quarrying activities by Grand Prosperous can lead to chance-finds. In the event that any artefact of cultural or historical value is found later, management will ensure that this discovery is reported to the National Heritage Conservation Commission for conservation.

6 POTENTIAL ENVIRONMENTAL AND SOCIAL IMPACTS

This section discusses the environmental and social impacts that are potentially likely to arise from the operation of the project. The section also discusses both the positive and negative impacts.

Table 6-1 below indicates the criteria used to characterise the various identified impacts. The significance of the impacts is discussed with regard to likelihood, duration, spatial extent and timing of occurrence. The discussion on the severity of the impacts also includes whether the impacts are reversible or not.

Table 6-1: Criteria and Terms used to describe Potential Environmental Impacts

Item no	Impact criterion	Description	Criterion classification	
			Term	Description
1	Positive or Negative Impact?	Will the impact have a positive or negative effect on the environment?	Positive	A positive impact.
			Negative	A negative impact.
2	Likelihood of Impact occurring?	What is the likelihood/ certainty associated with a potential impact?	Unlikely	Unlikely to occur.
			Possible	May possibly occur.
			Probable	Likely to occur.
			Certain	Certain to occur.
3	Timing of impact?	At what point in time will the impact occur?	Pre-project	Site construction phase.
			Start of project	Will occur immediately operations begin.
			Near-future	Will occur within the lifetime of the Project
			Mid-future	Will occur after the Project has closed.
			Distant-future	Will occur in the distant-future.
4	Duration of impact?	What is the likely duration or time over which the impact will occur?	Indeterminate	Unable to be predicted with certainty
			Short-term	Will cease once activity stops
			Medium-term	Will continue for the lifetime of the project.
			Long-term	Will continue beyond project closure
			Permanent	Will remain permanently.
5	Extent of impact?	What is the geographical extent of the impact?	Project Area	Will affect the immediate project area.
			Regional	Will affect areas outside the project area.
			Provincial	Will affect the Province.
			National	Will affect the whole of Zambia.
			International	Will affect other countries.
6	Significance of impact?	What is the severity of the impact, either positive or negative?	Very High	Very high impact.
			High	High impact.
			Moderate	Moderate impact.
			Low	Little impact.
7	Further studies?	Are further studies required to assess the significance of the impact?	1	Requires detailed assessment & specialist studies.
			2	Requires detailed assessment.
			3	Requires further assessment.
			4	Unlikely to require further assessment.
			5	Further assessment not required.

Table 6-2: Environmental/health risk assessment

ID No.	Impact	Positive or Negative	Sensitivity	Spatial Extent	Duration	Project Phase	Frequency	Likelihood	Reversibility	Further Studies
SOCIAL ECONOMIC IMPACTS										
1.	Contribute to national development by payment of taxes	Positive	High	National	Long term	Construction - Closure	Continuous	Certain	Yes	5
2.	Contribute to national development by offering employment opportunities	Positive	High	National	Long term	Construction - closure	Continuous	Certain	Yes	5
3.	Contribute to national development by offering business opportunities to suppliers and contractors	Positive	High	National	Long term	Construction - closure	Continuous	Certain	Yes	5
4.	Impact on land use due to the construction of the building	Negative	Low	Project Area	Long Term	Construction - closure	Continuous	Certain	Yes	5
5.	Impact posed by HIV/AIDS on public health	Negative	High	Provincial	Permanent	Construction - closure	Continuous	Possible	No	4
PHYSICAL ENVIRONMENT IMPACTS										
6.	Impact of workplace accidents on occupational health and safety	Negative	High	Provincial	Permanent	Construction - closure	Continuous	Possible	No	4
7.	Impact of dust emissions on air quality	Negative	Low	Project Area	Short term	Operation	Occasional	Probable	Yes	4
8.	Impact of vehicular emissions on air quality	Negative	Low	Project Area	Short term	Construction - Operation	Intermittent	Possible	Yes	5
9.	Impact on soil and water resources through disposal of waste	Negative	Low	Project Area	Short term	Construction - closure	Intermittent	Certain	Yes	5
10.	Impact of surface run-off on land and water resources	Negative	Moderate	Project Area	Short term	Construction - closure	Intermittent	Unlikely	No	5
11.	Impact of storm water on local drainage and the potential of overflowing	Negative	Moderate	Project Area	Medium term	Operation - closure	Intermittent	Unlikely	Yes	5
BIOLOGICAL ENVIRONMENT IMPACTS										
12.	Impact of ecological disturbance from land clearing	Negative	Low	Project Area	Short term	Construction	Occasional	Unlikely	No	5
13.	Erosion of top soil on disturbed areas	Negative	Low	Project Area	Short term	Construction	Occasional	Unlikely	Yes	5
14.	Impact of exploration activities on landscape and aesthetics	Negative	Low	Local	Long term	Construction - closure	Continuous	Certain	No	3
15.	Impact on air quality from noise generated by equipment and vehicles on birds	Negative	Low	Local	Short term	Construction - closure	Intermittent	Occasional	Yes	3
16.	Increase in traffic during the operation phase will affect	Negative	Regional	Local	Mid term	Construction - closure	Intermittent	Occasional	No	3
17.	Potential of fire risks on animal life	Negative	Low	Local	Short term	Operation	Occasional	Likely	No	2

There are a number of environmental and social issues associated with this development. The significant issues include:

6.1 POSITIVE IMPACTS

The following will be potential positive impacts to be derived from the project:

6.1.1 Economic benefits

The proposed project will likely bring development to Chirundu District which will have a multiplier effect on the economy of Zambia when a new mine is established due to this exploration activity.

6.1.1.1 Employment creation

Grand Prosperous intends to employ 25 people during the operation phase of the project. Employment opportunities make available income which in turn leads to the reduction in poverty levels.

6.1.1.2 Create business opportunities

A number of business opportunities will be created through the development of the proposed project. A number of suppliers will be on hand to supply personal protective wear, fuel, security, groceries to be sold in the shop, electrical and mechanical repair and maintenance of the vehicles.

6.1.1.3 Enhance local economy

Taxes, rates and pay-as-you earn remittances will be paid to, not only the central government, but also the local government in Chirundu and other surrounding districts. This will contribute to enhancing the local and nation's economy.

6.1.2 Ecological impacts

There are a number of abandoned pits from where illegal mining was previously done. Such pits are a danger to animals and people and aesthetically an eyesore. With Grand Prosperous being onsite and undertaking rehabilitation of such ecologically disturbed sites, there is expected to occur ecological restoration to such abandoned sites.

6.1.3 Physical impacts

Abandoned sites, which pose a danger to people, wildlife and livestock, will be buried and levelled.

6.2 NEGATIVE IMPACTS

6.2.1 Physical Impacts

The development of this project would lead to a number of negative environmental impacts as outlined below.

6.2.1.1 Impact on landscape and aesthetics

Exploration activities such as clearing for access routes and drill pads, grading and compacting using heavy machinery have potential to change the natural landscape and may therefore degrade areas of scenic beauty. Installation of camp structures may equally have an impact on the local aesthetics.

6.2.1.2 Impact on land use

As referred to under the chapter for baseline environment, the general land use is wooded vegetation. If sustainable mining practices are not put in place, existing land use can be negatively affected with a potential to affect ecological resources of the area.

6.2.1.3 Impact on water resources and surrounding property

Storage of fuel in bulk could lead to spillages that could pose a risk to the camp, adjoining property, the community and the environment. Poor handling of fuels may lead to spills which have a potential to contaminate groundwater resources.

6.2.1.4 Impact of surface run-off as a result of handling drill fluids

Diamond drilling involves the use of drill fluids which when spills and leaks occur have the potential to contaminate the soil and water resources.

6.2.1.5 Impact on soil texture due to use of heavy machinery

The use of heavy equipment such as rigs, vehicles and bulldozers, can compact and change the texture of the soil. This has the potential to leave the soil prone to erosion.

6.2.1.6 Contamination of groundwater via drilled holes

Poorly abandoned drill holes have the potential to contaminate groundwater by entry of poor-quality water and other foreign substances into the drill hole or through intermixing of waters from different aquifers. Unstable drill holes might also pose a physical hazard to people and wildlife.

6.2.1.7 Impact on quality of air quality due to emission of air pollutants

During site clearing for drill pads, access routes or camp establishment, dust will be generated and dispersed by the wind affecting a zone of up to 100m around the excavation site. Emission of loose

soil particles on cleared areas during strong winds may also affect the quality of the air in the immediate local environment.

Air pollutants are also likely to emanate from vehicular emissions due to increased vehicular traffic flows. However, this is negligible and transient.

6.2.1.8 Volatility of hydrocarbons

Hydrocarbons are volatile under certain conditions and their vapours in specific concentrations are flammable.

6.2.1.9 Impact of noise

Heavy machinery that will be used for the clearing and drilling activities are the likely sources of noise.

6.2.1.10 Impact on water and soil as a result of fuel handling

Soil contamination may result from poor handling of petroleum products such as oil and diesel during dispensing as well as improper disposal of waste oils, hydraulic fluids, and empty oil drums. Leaks from storage tanks can contaminate surface and subsurface soils, rendering drinking water from aquifers non-potable.

6.2.2 Biological Environmental Impacts

6.2.2.1 Impact on soil

Leaks from storage tank systems can have devastating environmental and economic impacts. Small quantities of diesel and gasoline can cause offensive odours, contaminate surface and subsurface soils, render drinking water from aquifers non-potable.

Storage and handling of fuel especially during transfer from one vessel to the other, can lead to spills. Such spills and leaks have a potential to contaminate the soil when mechanisms are not put in place to contain or treat the wastewater arising from spillages or leaks.

6.2.2.2 Generation of waste

Municipal solid waste such as kitchen waste, cans, plastic bottles, will be generated. Unsound management practices of waste have potential to pollute land and water resources.

6.2.2.3 Ecological disturbance caused by land clearing

Areas will need to be cleared for access routes, drill pads and establishment of camps. This activity has the potential to destabilise the existing natural ecosystems within the project area. The clearing of vegetation at drill sites and clearing of access routes is likely to affect the ecological value of the area. Some of these areas may be home to a diversity of plant and animal species. The normal growth of plant near drill pads and along the trails may be hampered by dust deposition from vehicles, although experience has indicated that this has negligible impact due in part to the deciduous nature of the majority of vegetation and the heavy rains in the wet season.

6.2.2.4 Introduction of invasive species

The access to the site either by foot or the use of vehicles can inadvertently introduce invasive species within the project site.

6.2.2.5 Erosion of cleared areas

Cleared areas will be exposed to adverse weather conditions such as rain and wind. The cleared areas will therefore be prone to soil erosion and may lead to siltation and sedimentation of nearby watercourses. The increase in turbidity is likely to have an impact on the quality of surface water.

6.2.2.6 Impact on land and water resources from wash water

The use of pit latrines as well as wash water from wash rooms and kitchen has the potential to pollute the soil and water resources.

6.2.2.7 Impact due to handling and storage of chemicals by contractors

Poor storage and handling of chemicals used can cause severe harm to employees through inhalation or exposure to the skin.

6.2.3 Social Economic Impacts

Implementation of this project will lead to the following socio-economic impacts

6.2.3.1 Impact posed by HIV/AIDS on public health

Due to increased income the increase of HIV/AIDS would increase in the local community.

6.2.3.2 Fire risks

Storage of fuel in large quantities poses a high risk of fire. This impact is significant and the effect of human life is irreversible.

6.2.3.3 Noise emissions

As alluded to above, heavy machinery such as operational vehicles and drilling machines emit elevated levels of noise that may interfere with hearing and generally the normal working conditions.

6.2.4 Occupational Health and Safety Impacts

6.2.4.1 Impact on public health

Pits left around the camp facility can provide a breeding ground for mosquitoes and inevitably lead to high incidences of malaria. In addition, pits could pose a potential physical hazard of injury in the event someone fell in especially for pits located close to working areas.

Where employees interact with the local people, taking into account improved earnings and coupled with unbecoming behaviour such as beer drinking, there is a possibility of the spread of HIV/AIDS.

6.2.4.2 Impact due to safety concerns

There are a number of operational activities that pose physical hazards and safety concerns for the workers. These include pits and trenches, bush fires, potential hazards posed by handling of fuel, use of drilling machinery and vehicles.

7 ENVIRONMENTAL, SOCIAL AND OCCUPATIONAL MITIGATION MEASURES

7.1 PHYSICAL ENVIRONMENTAL IMPACTS

The development of this project would lead to a number of environmental impacts as outlined below.

7.1.1 Impact on landscape and aesthetics

- Unless there is need for creating new routes, existing routes will be used for accessing project sites.
- Less well vegetated areas shall be preferred for temporary work camps.
- In order to minimise visual impact, the colour of tents shall blend in with the natural environment and the height of structures shall not extend beyond tree canopy.
- At the end of an exploration season, camps shall be dismantled and the camp area returned to its previous state as practicable, with thought given for the possible re-use in the following
 - field season.
- When camps are being decommissioned, all structures and garbage shall be removed from the site.
- Pits will be refilled with the excavated soil.

7.1.2 Impact on land use

- Make use of the existing tracks in the project area instead of creating new ones. Track clearing will only be carried out when it is absolutely necessary.
- Camps will be established at sites where the need for wholesale clearing is low.
- Arrange the camp in a circular fashion to minimize the risk of encroachment by animals. Arrange the camp so that there are no "dead ends" where wild animals may feel trapped if they enter the camp.
- Raise conservation awareness among employees and develop incident reporting procedures on poaching activities in the area.
- Collaborate with relevant stakeholders such as the Forestry Department, WARMA, and, local authorities in the conservation of natural resources in the project area.
- Undertake community engagement to explain to the local people the proposed activities, especially to the landowners where drilling will be carried out.

7.1.3 Impact on water and soil as a result of fuel handling

- To limit, contain, and manage the impact of spillages, fuel tanks/drums will be contained in a bund of sufficient capacity and will stand on an impervious surface.

- To limit the number of spills during transfer from drums to vehicles, hand pumps will be used.
- Fuel stores will be located on flat stable terrain at least 30 metres from the high-water mark of the nearest body of water whenever possible.
- Bunded areas shall be drained to a sump.
- The Geologist in Charge shall ensure that fuel containers are in good condition and shall be inspected for leaks, holes, dents and overall integrity.
- Waste fuel, oil and lubricants, soil contaminated by spills, used absorbent material, impervious linings and any waste contaminated by hydrocarbons (such as rags) will be treated as hazardous waste and disposed of appropriately. All hazardous waste oil will be stored onsite in dedicated drums for sale to recyclers.
- Where practicable a minimum 100-meters will be maintained between the drill site clearing and any water body or watercourse.

7.1.4 Impact of surface run-off as a result of handling drill fluids

- Only biodegradable mud and non-toxic additives shall be used.
- Drilling effluent shall be contained, in sumps, containers, or natural depressions located as close to the drill site as possible.
- The tank, container or earthen pit shall be constructed and maintained so as to prevent escape of any fluids there from, including any amounts that may be added by natural precipitation.
- At the completion of drilling, the sump is to be drained, the liner removed and the sump excavation filled in, with the subsoil replaced first and the topsoil replaced last.

7.1.5 Impact on soil texture due to use of heavy machinery

If the ground has been compacted by the use of heavy machinery or prolonged use, the drill pad shall be ripped to loosen the soil. Care shall be taken when ripping or loosening not to create large furrows which may become erosion gullies. This can be avoided by ripping along the contour.

Stockpiled soil and vegetation shall be spread over the site after activities have been completed.

The site shall be monitored periodically and further work carried out, if required, to ensure success of the rehabilitation.

Mulching will be carried out to accelerate seedling development and to reduce the chance of seed being washed away by rainfall and runoff. This practice can accelerate re-vegetation at higher-risk locations.

7.1.6 Contamination of groundwater via drilled holes

A drill hole abandonment procedure will be put in place and all exploration drill holes shall be abandoned as follows:

- Casing shall be removed from the drill hole to ensure placement of effective seals. If casing cannot be removed or has been cemented in position, the drillhole shall be securely capped at or below ground level.
- Install a pre-cast or in-situ cement or sodium bentonite plug to a minimum depth of 0.3m below ground level with plastic pipe breather to extend above ground level. Alternately install a plastic plug to a depth of 0.3m below ground level. Alternately completely backfill drillhole with cuttings to the surface allowing for settlement;
- The top 0.3m of fill shall consist of native soil, and a soil mound left over the hole's position to allow for subsidence.
- Place topsoil over the drillhole to encourage growth of natural vegetation. Complete to a height of 0.2m above surface to compensate for settlement.
- Measures shall be taken to ensure that sealing material does not migrate down or move up the drillhole under pressure during or after installation.
- All drill pads, mud pits, and other disturbances shall re-contoured and stabilized.
- Where drill holes have intersected artesian water, they shall be capped below ground surface and that major aquifers intersected down hole shall be sealed off.

7.1.7 Impact on air quality due to emissions of air pollutants

To reduce the impact of air pollutants the following measures will be undertaken:

- In the event of excess dust being generated, personal protective equipment (dust masks) will be used.
- Employ appropriate measures to reduce dust by:
 - Keeping disturbed areas to a minimum, and re-vegetating bare areas, as soon as possible;
 - Minimising vehicle movements and speed.

7.1.8 Impact of noise

Where workers are exposed to elevated sound levels, use of personal protective equipment such as ear muffs/plugs is mandatory.

To reduce the impact of noise on occupational health, all equipment will be regularly maintained to ensure that it is mechanically sound.

7.1.9 Volatility of hydrocarbons

Appropriate warning signs posts will be erected near the fuel storage area and no naked fires will be allowed near the fuel drums. No smoking will be allowed anywhere the site. Sand buckets will be provided alongside booms to contain any spill or leaks that can potentially catch fire.

7.2 BIOLOGICAL ENVIRONMENTAL IMPACTS

7.2.1 Erosion of cleared areas

Where topsoil is removed during drill pad preparation, this will be stockpiled for use when rehabilitating the site. At the completion of drilling, the topsoil shall be re-spread and, where appropriate, the area re-vegetated with species consistent with the surrounding vegetation.

Cutting of trees will be limited so as to provide a windshield against soil erosion.

Slash material is to be stockpiled at the edge of the clearing and utilized for reclamation of the site.

7.2.2 Ecological disturbance caused by land clearing

- When using or constructing tracks, trimming of overhanging timber will be kept to a minimum.
- Trails will only to be cleared to the width of the vehicles travelling the road.
- Clearing for drill pads will depend and be limited to the size of the drill rig. Clearing shall be kept to a minimum size. A standard drill site should be constructed to facilitate drilling operations and shall not exceed 20 meters by 20 meters (or 400 square meters).
- Camps will be located on previously cleared sites or areas where other land use possibilities are low.
- Rehabilitation activities such as soil re-profiling, mulching and in some cases, re-vegetating using indigenous plant species will be practiced.
- Speed limit will be established and Grand Prosperous will ensure that drivers do not exceed the set speed limit.

7.2.3 Generation of waste

- All solid wastes shall be removed from the camp to an approved waste disposal site.
- No burying or burning of wastes shall be permitted.
- Waste shall be stored in such a manner that it does not generate nuisances of odour and unsightly visual impact. For storage, solid waste will be located in covered, leak proof containers. Food waste shall be kept in covered, fly/animal proof containers until removed to an approved waste disposal site.
- The camp and drill sites shall at all times be kept in a safe, neat, and sanitary condition.

- Any hazardous waste generated onsite shall be handled differently. All hazardous waste will be kept on a sheltered concrete. Lead-acid batteries from vehicles shall be sold off to recyclers for the reclamation of lead and plastic. Waste oil will be kept in sealed drums before being sold to waste oil recyclers. No hazardous waste oil will be burnt or buried.

7.2.4 Impact on land and water resources from wash water

- Pit latrines shall be located at least 100m from the nearest body of water, topographically lower and downwind from the camp, and at least 30 to 40 metres away from the kitchen area. To promote personal hygiene, handwashing facilities will be located between the latrine and camp.
- Any sump or pit used for storing liquid waste shall be located a minimum of 100 metres back from the high-water mark of any water body or watercourse; and 15 metres away from drilled wells. There shall be a minimum of 2 meters of isolated distance between the bottom of the pit and the water table where the separation material is clay and 5 meters where it is sand.
- Wash water from the kitchen and washing facilities shall be discharged to the sumps located at least 100 metres from any body of water. The bottom of the pit shall be filled with coarse gravel and the sides shored up with board, etc. to prevent erosion and collapse of the pit.
- Wastewater shall not be discharged directly into a body of water.

7.2.5 Impact due to handling and storage of chemicals by contractors

- Only biodegradable chemicals will be used.
- Provide adequate ventilation in the storage area to prevent build-up of vapour or aerosols;
- Containers will be inspected for leaks or spills, rust and high temperature on a regular basis;
- All containers will be carefully secured, clearly identified and stored.
- There shall be no smoking in the area where chemicals are being handled or stored.
- In an event of a major spillage any unauthorized persons will be prevented from accessing the area.
- Any spillage shall be contained with appropriate absorbent materials to prevent the chemicals from reaching bodies of water.
- All chemicals will be arranged according to their respective categories and accompanied with relevant Material Safety Data Sheets (MSDS);
- Storage facilities will be posted with the appropriate signage so as to warn people of the danger associated with chemicals.

- The store room will be bunded with concrete slabs and all floors will have an impermeable surface.
- All chemicals in the storehouse will be placed on pallets.
- First aid equipment will be provided.

7.2.6 Impact on water resources due to abstraction of water

Careful use of water resources will be implemented to reduce wastage arising from spills and leaks. Awareness will be created among the work force on the need to conserve water. Recycling of water during drilling will be instigated.

7.2.7 Introduction of invasive species

Grand Prosperous will be carrying out a biodiversity survey, on a yearly basis, to determine if there are any invasive species introduced in the project. No planting of any ornamental tree will be allowed within the project site.

7.3 SOCIAL ECONOMIC IMPACTS

Implementation of this project will involve the use of both skilled and unskilled labour from the construction phase up to operation phase.

The project has employed 50 staff and the project will involve the use of geologists, technicians, mechanics, drivers, accountants, and cooks. Provision of employment contributes to raising the socio- economic well-being of the people and thereby reducing poverty levels.

Management intends to employ local people as far as possible unless the required expertise cannot be found locally.

7.3.1 Paying of taxes

The operation of this project will contribute to enhancing the nation's economy through paying of taxes in form of duty and pay as you earn remittances.

7.3.2 Contracting of service suppliers

A number of service suppliers will be contracted to supply services such as food, transport, fuel, and drilling operations.

7.4 OCCUPATIONAL HEALTH AND SAFETY IMPACTS

7.4.1 Impact on public health

To reduce the incidence of malaria, any standing water will not be allowed to accumulate in ponds or pits. All containers and equipment where water can collect and provide breeding places will be removed.

Sensitization campaigns on HIV/AIDS will be carried out among employees.

Danger signs will be erected around the pits while re-profiling and re-vegetating will follow.

7.4.2 Impact due to safety concerns

- Field crew shall be trained in firefighting techniques and the use of firefighting equipment.
- Firefighting equipment shall be visible and the location easily accessible.
- Locate the kitchen area at least 20 metres downwind from sleeping tents. Locate tents, kitchen area, and fuel storage area with fire prevention in mind.
- Firebreaks shall be constructed around the camps. The firebreak shall be at least 10 metres wide.
- Fire drills shall be held regularly.
- Flammable materials shall be stored away from ignition sources.
- There shall be dedicated fire-fighting equipment fitted to all exploration vehicles and machinery. All heavy equipment and fuelling sites shall have approved and fully charged fire extinguishers installed.
- All fuelling sites shall be designated and no smoking allowed near fuelling sites or while operating equipment.
- Clear brush and grasses from around portable generators, water pumps, etc.
- All pits shall be cordoned off with reflective material.

8 ENVIRONMENTAL MANAGEMENT AND MONITORING PLAN (EMMP)

The primary goals of the environmental management and monitoring system will be:

- (a) To avoid the contamination of the environment and negative ecological impacts
- (b) To enhance compliance to environmental laws and regulations

Overall responsibility will be that of the project who will be required to ensure that the project and contractors comply with health, environmental and safety standards.

Table 8-1: Environmental management and monitoring plan

Aspect	Impact	ID No.	Mitigation/Enhancement Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Responsibility	Cost
Service suppliers	The use of local service suppliers increases their competence and maximises their ability to carry out related works	1.	Management will use companies registered in Zambia for the provision of services such as, transport, consumables, general consultancy, etc	Quarterly	Throughout the project stages	Register of local service suppliers	Director	No Cost
Local employment	Ensure that the locals are offered first priority during the recruitment exercise	2.	Ensure that employment opportunities are offered to locals, except for jobs that require specialisation	Quarterly	Throughout the project stages	Register of local employment	Director	No cost
		3.	Ensure that the right balance with regard to the local employment and gender is observed.	Monthly	Installation	Register of local employment	Director	No cost
		4.	Clearing will be restricted to the exploration site	Daily	Construction	Area cleared	Director	No cost
Land clearing	Compacting of soil by use of heavy machinery	5.	All design and installation work to be carried out by contractors who are qualified and accredited for this purpose.	Weekly	Construction	List of accredited contractors	Director	No cost
Generation of vegetative waste	Soil/land pollution after disposal	6.	Ensure waste is disposed of at designated site	Weekly	Construction	Clean project site	Contractor	K200/week
Emission of dust	Air pollution	7.	Watering of the cleared areas	Daily	Construction	Register of employees	Contractor	K250/week
Engagement of casuals for work	Improved way of life	8.	Engage casuals from nearby places	Monthly	Construction	List of employees	Contractor	No cost
Fuel handling and storage	Spills and leaks	9.	- use of corrosion resistant material - regular monitoring to detect the likelihood or presence of leaks - staff training on leak detection and prevention	Monthly	Operation	Specs of materials used and training records	Director	Investment cost
	Land contamination	10.	Apply preventative techniques to prevent leaks and spills	Monthly	Construction -operation	Absence of oil contaminated land	Contractor/ Site Manager	Operation cost
Generation of used oil	Land contamination	11.	Storing of used oil in easily identified drums	Monthly	Construction -operation	Absence of oil contaminated land	Site Manager	K1000

Aspect	Impact	ID No.	Mitigation/Enhancement Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Responsibility	Cost
Emission of dust	Air pollution	12.	Observe speed limits in dusty areas	Daily	Construction	Dust free ambient air	Contractor	No cost
Generation of noise	Noise pollution	13.	Ensure vehicles are in sound mechanical condition	Daily	Throughout project life	Absence of vehicle-related noise	Contractor/ Site Manager	K5000/ month
Driving of vehicles	Increased risks of accidents	14.	Observing traffic rules and speed limits	Daily	Throughout project life	Absence of vehicle-related accidents	Contractor/ Site Manager	No cost
		15.	Training of drivers in safety issues and emergency preparedness	Yearly	Throughout project life	Training records	Contractor/ Station Manager	K2,500
Spills and leaks	Land contamination	16.	- staff training on leak detection and prevention - spill kits in place	Quarterly	Operational area	Absence of leaked/spilled oils	Site Manager	K50,000
	Risk of fire	17.	Emergency preparedness	Quarterly	Operation stage	Availability of plans and tools to deal with emergencies	Site Manager	K3,000
		18.	Post warning signs against smoking, having naked lights and use of the phone	Bi-annually	Operation stage	Availability of warning signs	Site Manager	K500
Minor evaporation when filling tanks	Emission of greenhouse gases	19.	- No smoking in the area where fuels are being handled or stored shall be allowed. - Train employees in sound management of fuels and waste	Bi-annually	Operation stage	Absence of emergencies	Site Manager	K5000
Land use	Reduce significant land use changes as a result of land levelling	20.	Uphold best environmental management practice aimed at protecting the areas' aesthetics	Weekly	Construction	Absence of penalties and/or complaints from neighbours	Director	Operational cost
HIV/AIDS	Prevent or reduce the impact of HIV/AIDS	21.	Educate and sensitize workers on the dangers of HIV/AIDS and promote self-protection, abstinence from casual sex and the use of condoms	Monthly	Throughout the project stages	Knowledge of HIV/AIDS among workforce	Director	K5,000
		22.	Use local employment unless the required expertise cannot be obtained locally	Monthly	Throughout the project stages	Register of local employment	Director	No cost

Aspect	Impact	ID No.	Mitigation/Enhancement Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Responsibility	Cost
Generation of waste	Prevent impact of waste on public health due to inappropriate waste management practices	23.	Ensure that no littering occurs at exploration sites	Daily	Throughout the project stages	Absence of penalties and/or complaints from neighbours	Supervisors	No cost
		24.	Reuse and recycle materials from the process	Quarterly	Operation	Percentage of recycled materials	Supervisors	No cost
		25.	All solid wastes will be stored in designated receptacles on site and removed from the site for disposal.	Daily	Throughout the project phase	Number of sturdy waste receptacles	Supervisors	K4,000
		26.	Maintain all equipment and machinery in a sound mechanical state	Weekly	Throughout the project stages	Absence of elevated noise	Director	Operational cost
Noise generation	To reduce the impact of noise on occupation health and surrounding environs	27.	Provide all workers who are exposed to elevated noise with ear muffs	Daily	Throughout the project stages	Absence of elevated noise	Director	K6,000
		28.	Restrict working times to day-time only	Daily	Throughout the project stages	Absence of elevated noise	Director	No cost
		29.	Use personal protective equipment (dust masks) in the event of excess dust being generated	Daily	Throughout the project stages	Absence of penalties and/or complaints from workers	All Supervisors	K8,000
		30.	Ensure that adequate OHS measures and protocols are put in place, made available to workers and adhered to.	Quarterly	Throughout the project stages	Absence of penalties and/or complaints from workers	Director	Operational costs
Air Quality	Prevent localised air pollution due to emission of dust	31.	Implement regular training in OHS. Ensure that these form daily safety talks before beginning works	Quarterly	Throughout the project stages	Absence of injuries among workforce	Supervisor	Operational cost
		32.	Provide appropriate and adequate PPE for the workers	Daily	Throughout the project stages	Absence of injuries among workforce	Director	K45,000
		33.	Provide means of escape in case of fire	Annually	Construction	Presence of fire exists	Site Manager	Included in investment cost
		34.	Provide adequate fire-fighting measures at the premises which will	Quarterly	Throughout the project stages	Number of fire-fighting	Site Manager	K25,000
Fire hazards	Prevent the risks associated with fire at the work place							

Aspect	Impact	ID No.	Mitigation/Enhancement Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Responsibility	Cost
			include extinguishers, hydrants, hose reels, automatic warning			equipment on site		

9 EMERGENCY CONTINGENCY PLAN

The emergency contingency plan is a supplementary to the Environmental Management Plan. The emergency contingency plan describes the procedure to be followed by the employees in case of an emergency. The Project Manager appointed Safety officer and all line officers will be responsible for the implementation of the Emergency Contingency. Table 9-1 below presents the Emergency Response Plan.

Table 9-1. Emergency Response Plan

Emergency Situation	Cause	Proposed Response	Respondents
Staff Injury	- Unskilled labour - Neglect of safety procedures - Faulty equipment and tools - Sheer accident	- Apply appropriate First Aid - Document incidence - Take to hospital if necessary - Investigate cause and institute appropriate measures to prevent similar occurrences	Key Respondent: Safety officer Other Respondents: First Aid Attendant on Duty, Immediate supervisor, Project Manager, Hospital staff, Zambia Police
Fire outbreak	- Neglect of safety procedures - Faulty electrical connections - Carelessness e.g smoking in non-smoking areas	- Sound alarm and instruct all to assemble at fire Assembly point - Conduct roll call - Fight the fire using appropriate tools (fire extinguisher, sand, water) - Chirundu Town Council - Isolate area with barrier tape - Clean-up site - Document incidence	Key Respondent: Safety officer Other Respondents: - Emergency Response Team - Zambia Police - Chirundu Town Council

10 DECOMMISSIONING AND CLOSURE

Exploration closure planning primarily considers the best way to manage environmental changes and socio-economic effects associated with closing the project. Successful closure generally varies from place to place due to several factors. However, it is important that physical, chemical and biological stability is achieved and the final land use is appropriate. Messrs Grand Prosperous have budgeted US\$15,000 to meet project closure costs. Table 10-1 presents the summary of closure activities and estimated costs.

Table 10-1. Summary of the closure activities and estimated costs

Project Component	Closure Activity	Estimated Cost (US\$)
Geological Mapping	Nil	Nil
Induced Polarisation & Electromagnetics	Nil	Nil
Exploration and Sampling Pits and Trenches	All pits and trenches shall be stabilized and then buried with a blocking tape put around them and warning signs erected	7,500
Percussion Drilling	- Burying drilling mud immediately each hole is completed - plugging the drill hole with concrete	2,500
Core (Diamond) Drilling	- Burying drilling mud immediately each hole is completed - plugging the drill hole with concrete	2,500
Camp Site	- Dismantling all the tents and other structures (there will be no permanent structures) - Disinfect portable toilets - Re-vegetate - Close access roads to avoid further deforestation activities since some areas may become more accessible - Lime shall be used to disinfect portable toilets, camp houses shall be demolished and the sight rehabilitated.	2,500
Total Estimated Closure Costs		15,000

11 CONCLUSION

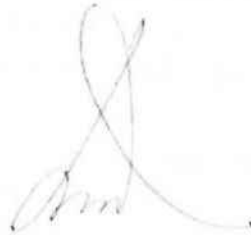
The development of this project will undoubtedly bring economic development and create other business opportunities. Management has considered all significant potential negative environmental impacts likely to arise from this undertaking and has proposed mitigating measures. All relevant pieces of legislation will be adhered to so as to enhance environmentally sound practices.

The major focus of the environmental management and monitoring plan will be to ensure that spill and leaks are prevented and, in the event, that they occurred, contain them. Maintaining safety standards will occupy the daily programme of the Project Manager. To ensure that these SHE standards are observed, management will train employees in leak and spill detection, environmental and safety management.

It is for this reason that we propose that the project be granted positive consideration in view of the accumulation of positive results likely to result from the project.

12 DECLARATION OF AUTHENTICITY OF REPORT CONTENTS

I wish to certify that this Environmental Project Brief was prepared consistent with the level of effort involved based on the information available at the time of preparation and the data supplied by outside sources. This EPB conforms to the requirements of the Environmental Impact Assessment Regulations, SI 28 of 1997 with regard to the development of Environmental and Social Management Plans.



Director – Grand Prosperous Company Limited

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APPENDICES

APPENDIX 1: CONSENT LETTER FROM HRH CHIEFTAINNESS SIKOONGO



REPUBLIC OF ZAMBIA
HRH Chieftainess Sikoongo's Own Palace

Sikoongo Main & Lusitu
Sub-Court
P.O. Box 30
Chirundu

Your Ref:

Our Ref:

25TH July, 2024

The Director
GRAND PROSPEROUS COMPANY LIMITED,
Lusaka – Zambia

Dear Sir,

**RE: GRANT OF CONSENT TO CONDUCT EXPLORATION OF GYPSUM TO
GRAND PROSPEROUS COMPANY LIMITED IN SIKOONGO CHIEFDOM
CHIRUNDU ZONE AREA IN CHIRUNDU DISTRICT SOUTHERN
PROVINCE.**

Reference is made to the above subject matter, with regards to your application dated 25th February 2024, where you applied to the chieftdom to undertake Exploration of Gypsum in Chirundu Zone in Sikoongo Chieftdom.

The Sikoongo Chieftdom is pleased to **Grant Consent** to your company Grand Prosperous Company Limited to carry out the Exploration of Gypsum within Chirundu Zone of Chirundu District, within the requirements of the Mines and Minerals Development Act. No 11 of 2015, and other relevant legislation.

Your exploration and mining works should give preference to the Sikoongo Chieftdom subjects in terms of jobs and other social –economic benefits.

Yours In Traditional Services.


Karen Beatrice Kajembwe

HRH Chieftainess Sikoongo

CC. Traditional Council Sikoongo



Wards: Sikoongo, Chirundu, Ibbwe-munyama, Ingombe-ilede, Lusitu, Musaya, Nanyunga, Kapululira, Kanyanga, Mushunga, Nkandabwe
Promoting & Protecting our Traditions & Cultural Heritage

APPENDIX 2: EXPLORATION LICENSE FOR GRAND PROSPEROUS COMPANY LIMITED

Siavonga



REPUBLIC OF ZAMBIA

The Mines and Minerals Development Act, 2015
(Act No. 11 of 2015)
The Mines and Minerals Development (General) Regulations, 2016

LICENCE NO. 36411-HQ-LEL

LARGE SCALE EXPLORATION LICENCE
(Section 23 of the Mines and Minerals Development Act, No. 11 of 2015)

Holder's name: Grand Prosperous Company Limited

Address: Plot 2120, Ndola Road, Dallas, Kabwe

The Prospecting area shall be the area described in the Schedule and annexed hereto and bordered Blue on the Plan.

The licence relates to the following minerals: Gypsum, Anhydrite, Lead, Quartz, Fluorine, Fluorite, Talc-Soapstone, Feldspar, Granite, Clay (General), Coal and Zinc

The licence is granted for a period of 4 years commencing on the 27th day of June, 2024

The conditions of grant of the licence are as shown in the Annexures attached hereto.

Issued at Lusaka this 11th day of July, 2024.


Samuel C. Maango
Director

ENDORSEMENT OF REGISTRATION
This Large Scale Exploration Licence has on this 27th day of June, 2024 been registered in the Register.


Samuel C. Maango
Director

APPENDIX 3: COMPANY CERTIFICATE OF INCORPORATION

Companies Form 7
Companies Registration No. 120150135131
Serial No. 1001712



Republic Of Zambia

CERTIFICATE OF INCORPORATION OF
COMPANY LIMITED BY SHARES

(Section 10)

This is to certify that GRAND PROSPEROUS COMPANY LIMITED is on and from the
26th day of August 2015 incorporated as a COMPANY LIMITED BY SHARES.

Given under my hand and seal at Lusaka, Zambia, this 26th day of August 2015.



N.J. Moola
Assistant Registrar of Companies

For further details relating to this business visit
<http://www.pacra.org.zm>

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